



US012404622B2

(12) **United States Patent**
Kim

(10) **Patent No.: US 12,404,622 B2**

(45) **Date of Patent: Sep. 2, 2025**

(54) **CLOTHES TREATING APPARATUS**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 372 days.

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(21) Appl. No.: **17/992,497**

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(22) Filed: **Nov. 22, 2022**

International Search Report dated Jan. 25, 2023 issue in PCT Application No. PCT/KR2022/014577.

(65) **Prior Publication Data**

US 2023/0212808 A1 Jul. 6, 2023

(Continued)

Related U.S. Application Data

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(63) Continuation of application No. PCT/KR2022/014577, filed on Sep. 28, 2022.

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(30) **Foreign Application Priority Data**

Jan. 3, 2022 (KR) 10-2022-0000244

(57) **ABSTRACT**

(51) **Int. Cl.**
D06F 39/08 (2006.01)
D06F 39/02 (2006.01)
(Continued)

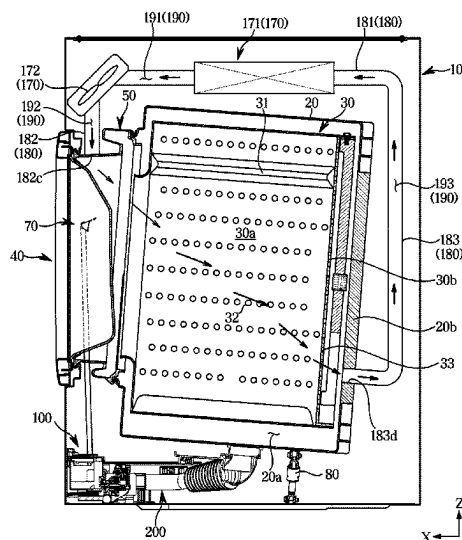
A clothes treating apparatus including: a cabinet; a tub provided in the cabinet and configured to collect wash water; a circulation device configured to circulate the wash water contained in the tub and including a circulation pump, a first circulation path to guide the wash water from the tub to the circulation pump, and a second circulation path to guide the wash water from the circulation pump to the tub so that the wash water circulates by flowing from the tub to the circulation pump through the first circulation path and flowing back to the tub from the circulation pump through the second circulation path; and a detergent supply device configured to supply a detergent to the circulation device and including a detergent case, and a detergent pump connected to the detergent case and mounted on the first circulation path to supply the stored detergent through the first circulation path.

(52) **U.S. Cl.**
CPC **D06F 39/085** (2013.01); **D06F 39/022** (2013.01); **D06F 39/087** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC D06F 39/085; D06F 39/022; D06F 39/087;
D06F 39/088; D06F 39/06; D06F 39/10;
D06F 2105/06; D06F 2105/42

See application file for complete search history.

20 Claims, 14 Drawing Sheets



- (51) **Int. Cl.**
D06F 39/06 (2006.01)
D06F 39/10 (2006.01)
D06F 105/06 (2020.01)
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- (52) **U.S. Cl.**
CPC *D06F 39/088* (2013.01); *D06F 39/06*
(2013.01); *D06F 39/10* (2013.01); *D06F*
2105/06 (2020.02); *D06F 2105/08* (2020.02);
D06F 2105/42 (2020.02)

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FIG. 1

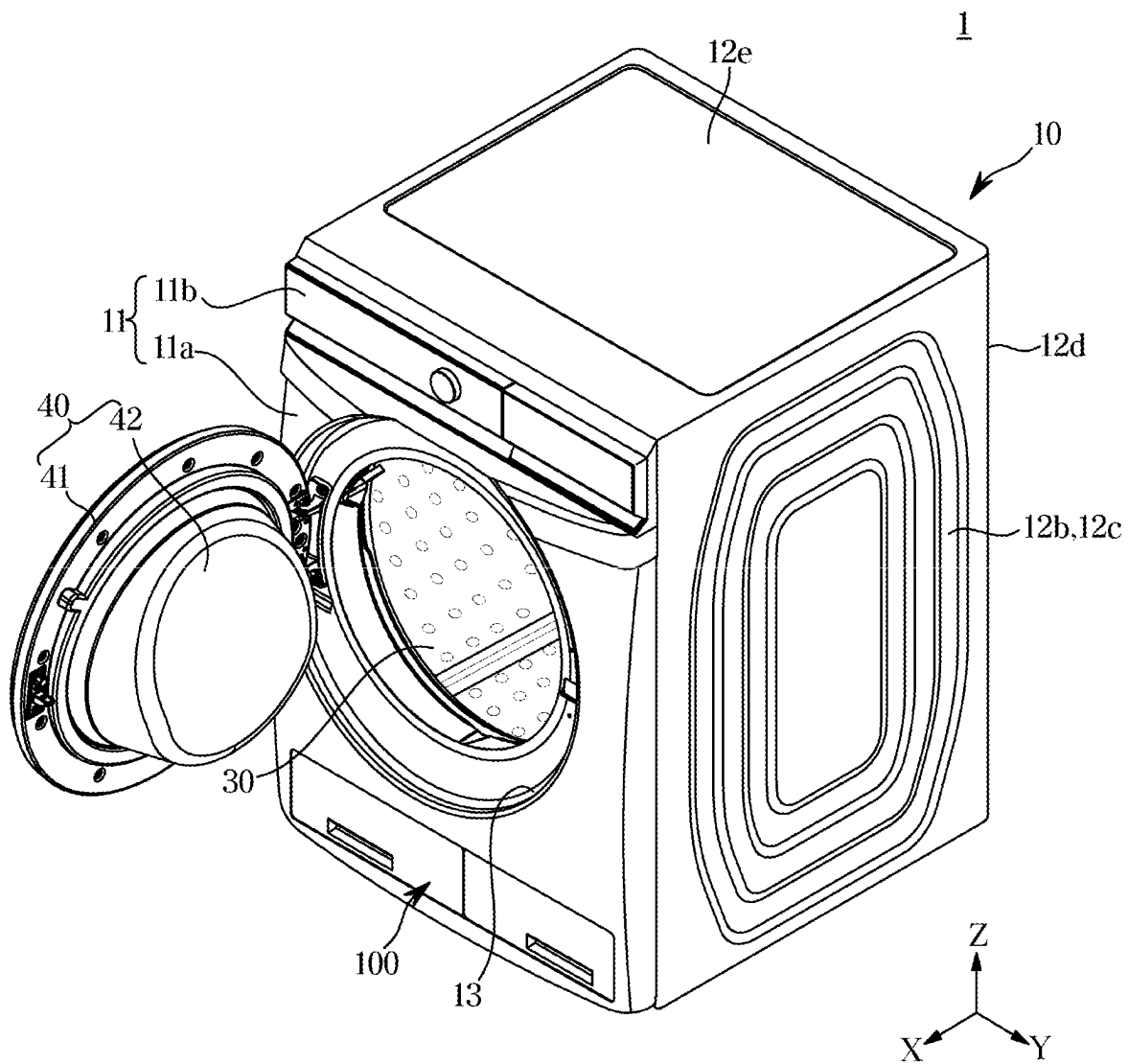


FIG. 2

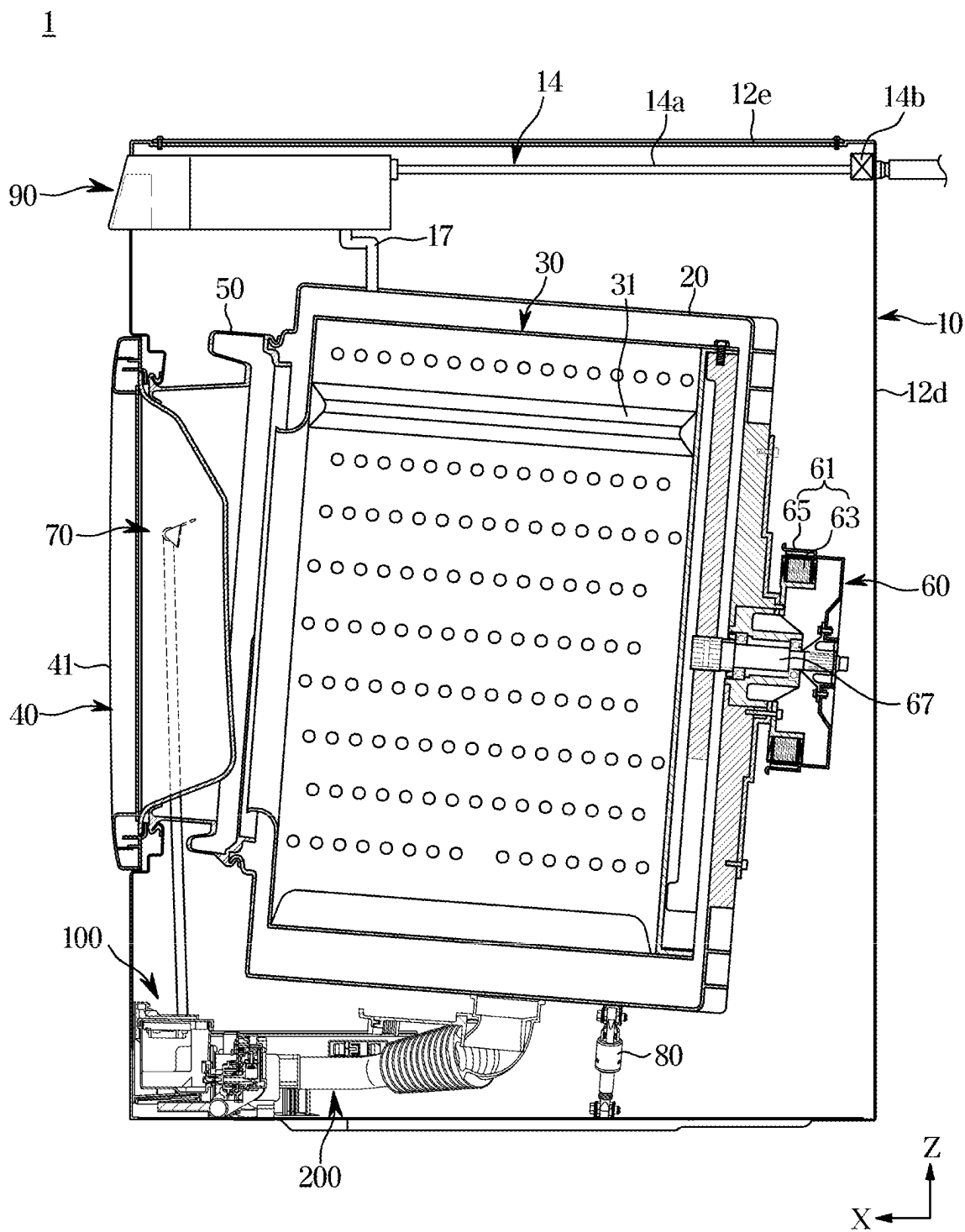


FIG. 3

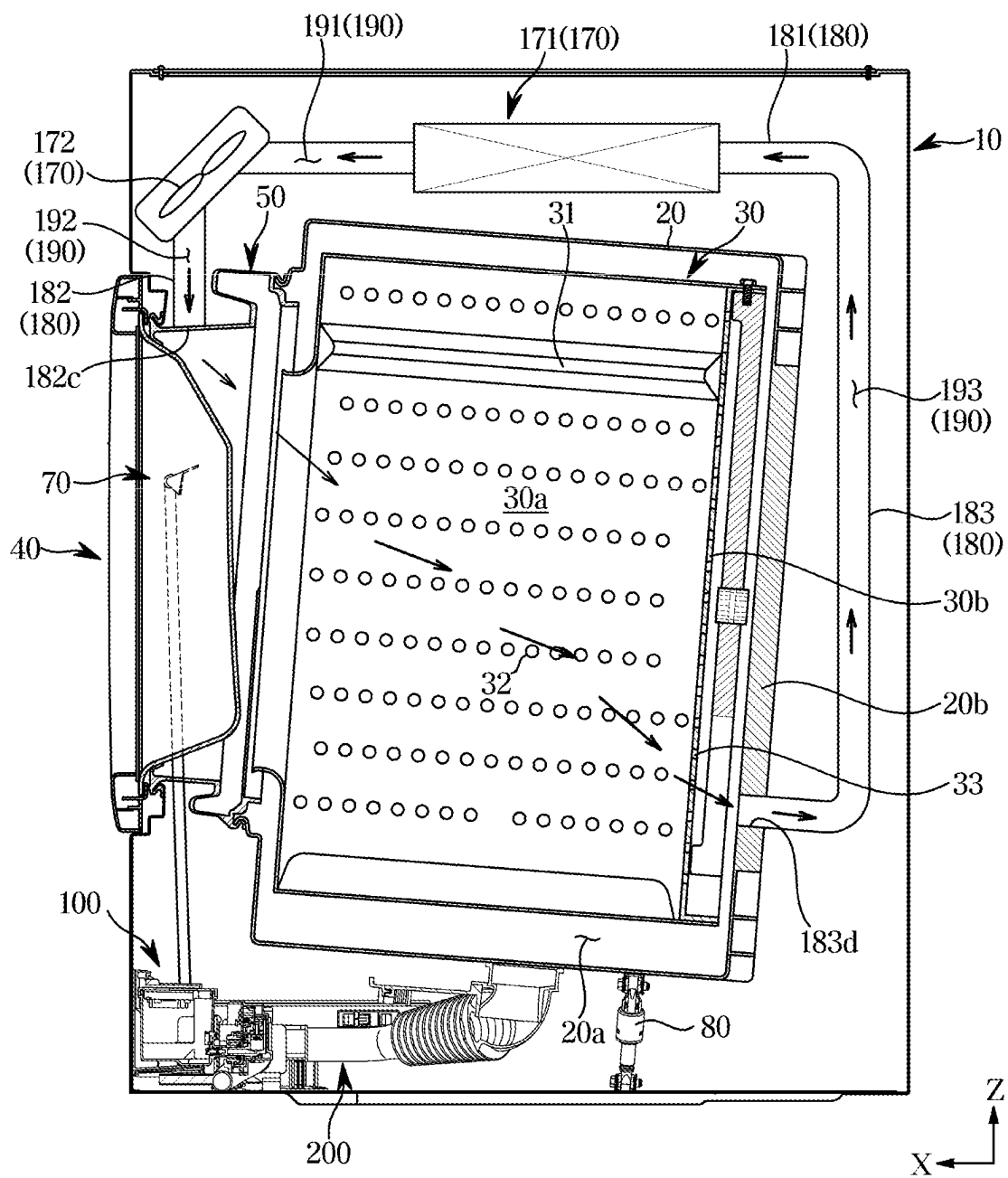


FIG. 4

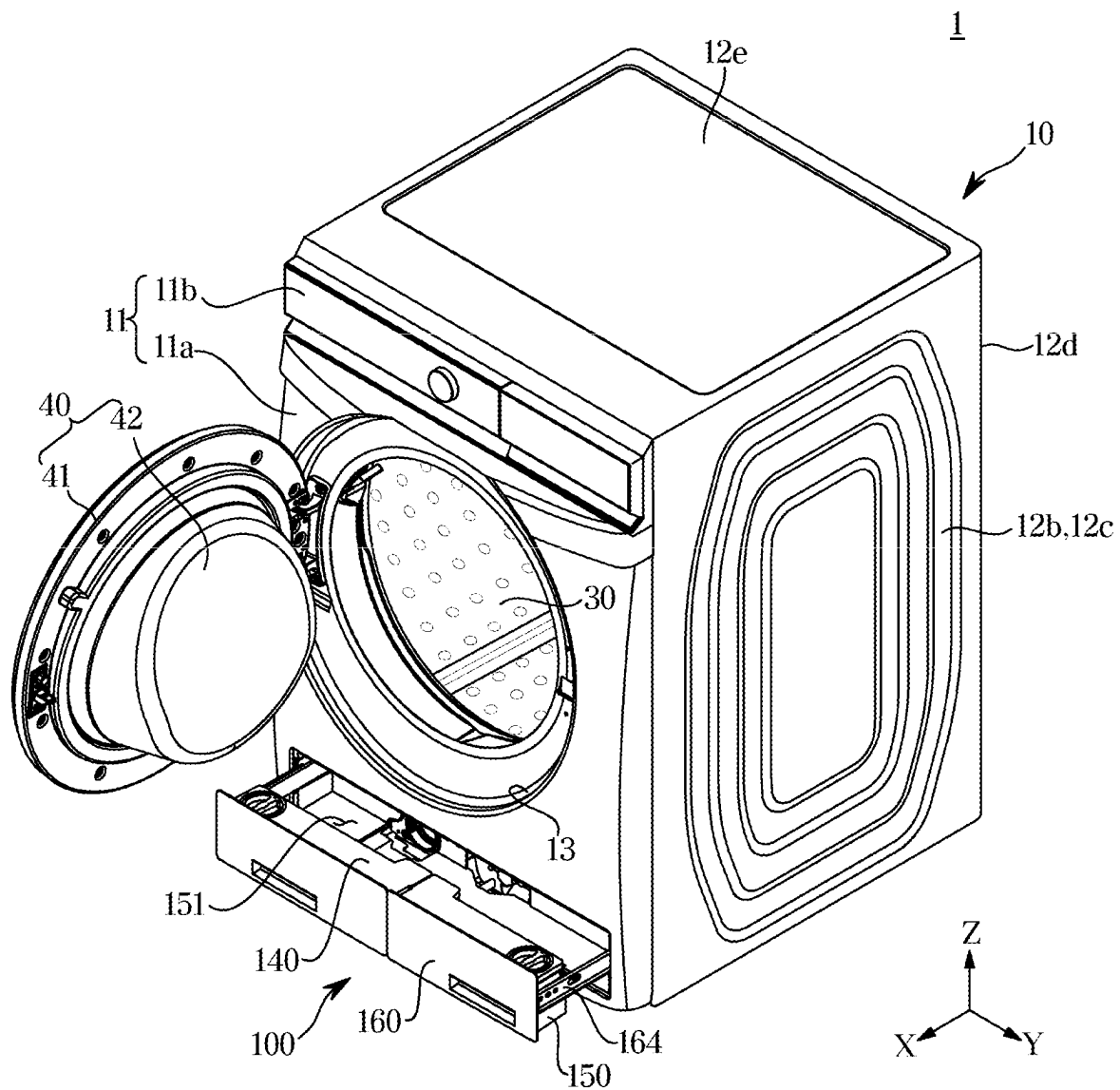


FIG. 5

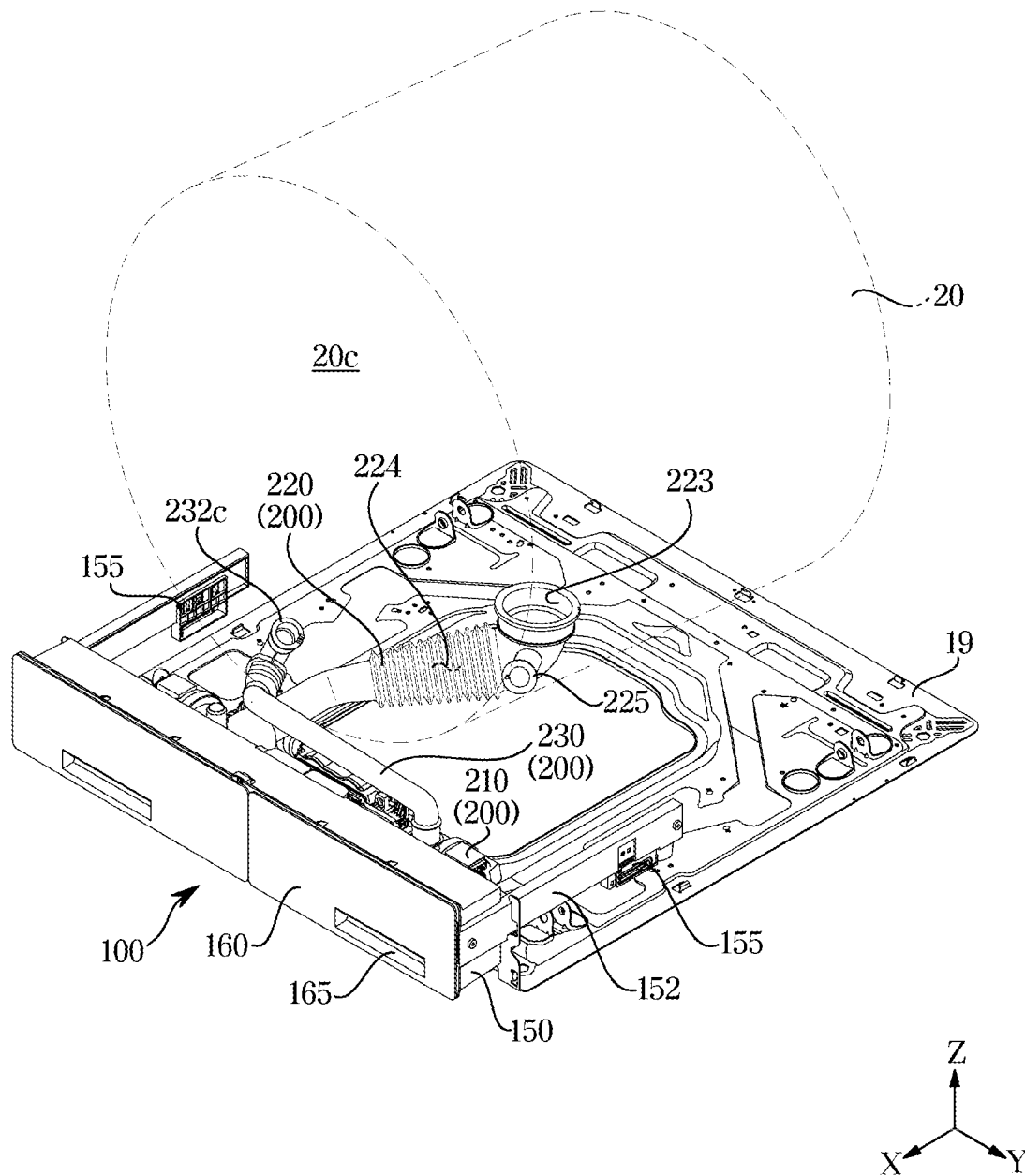


FIG. 6

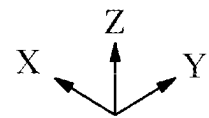
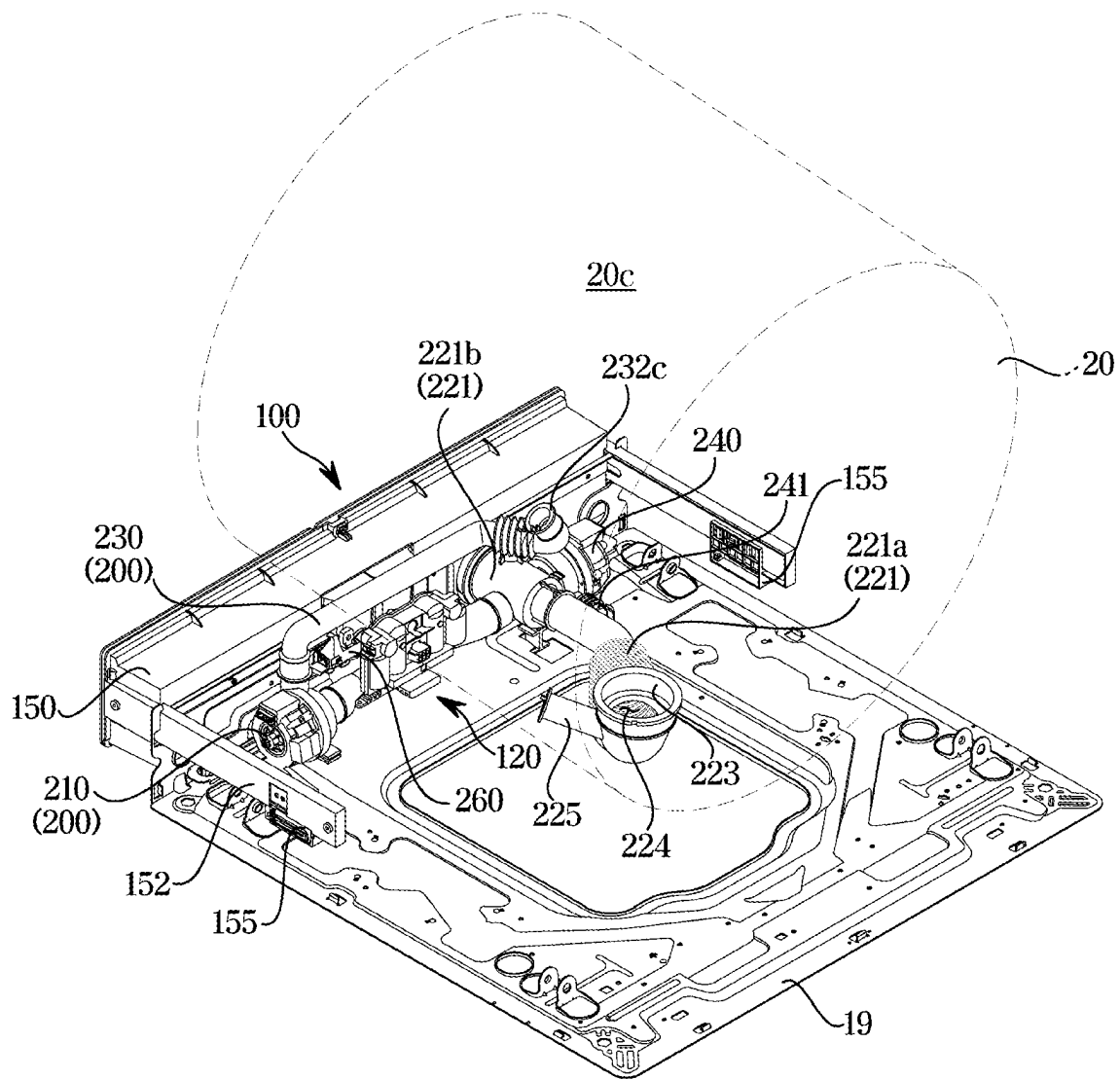


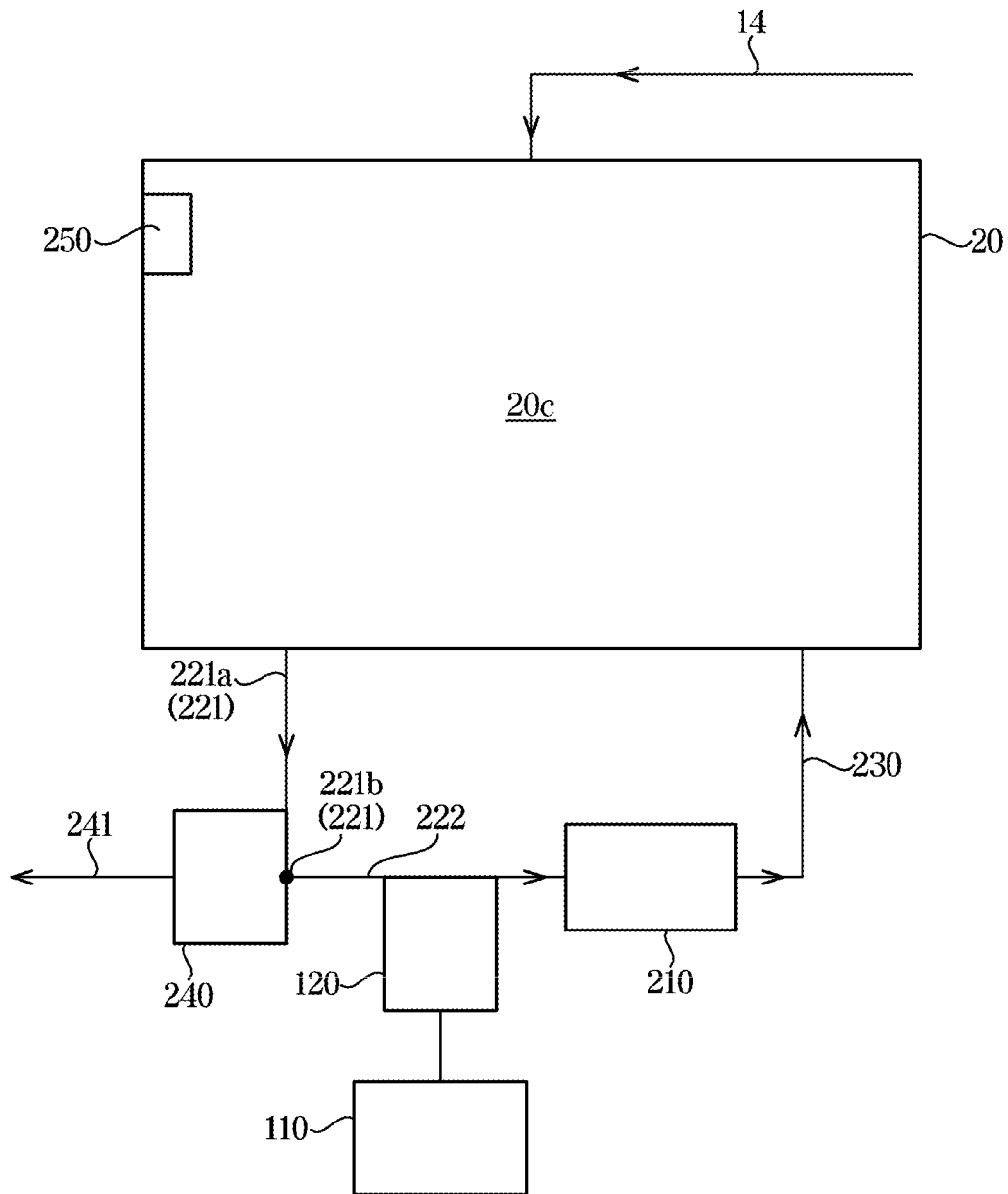
FIG. 7

FIG. 8

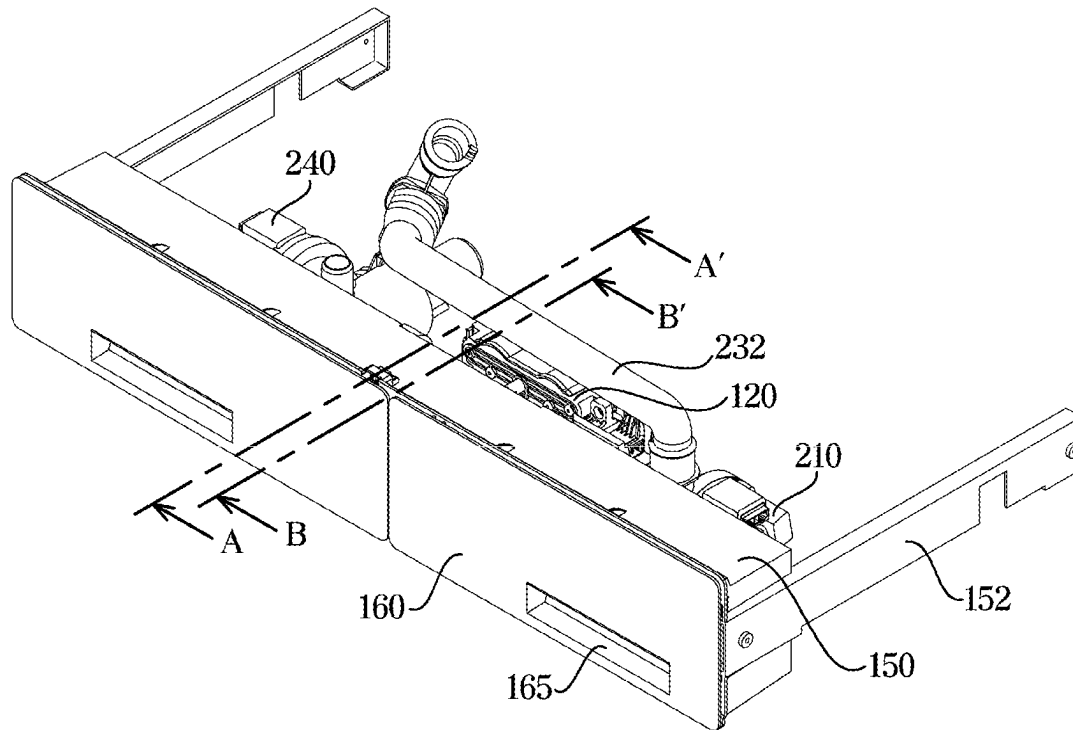


FIG. 9

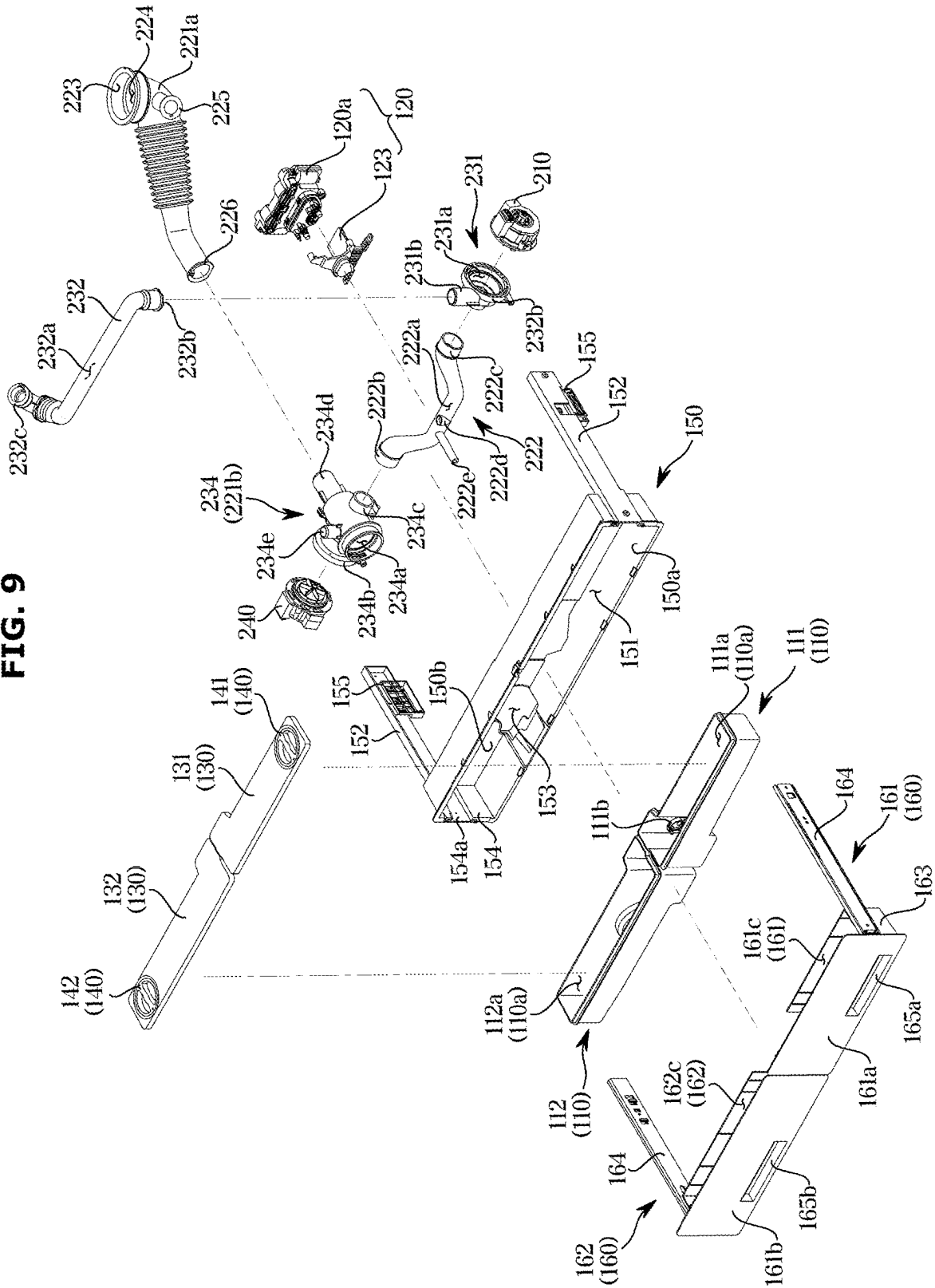


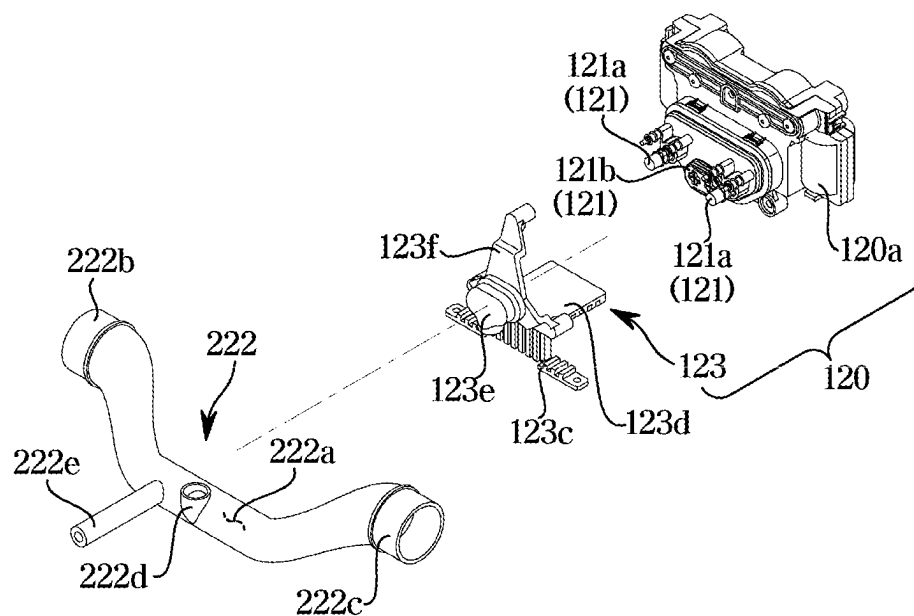
FIG. 10

FIG. 11

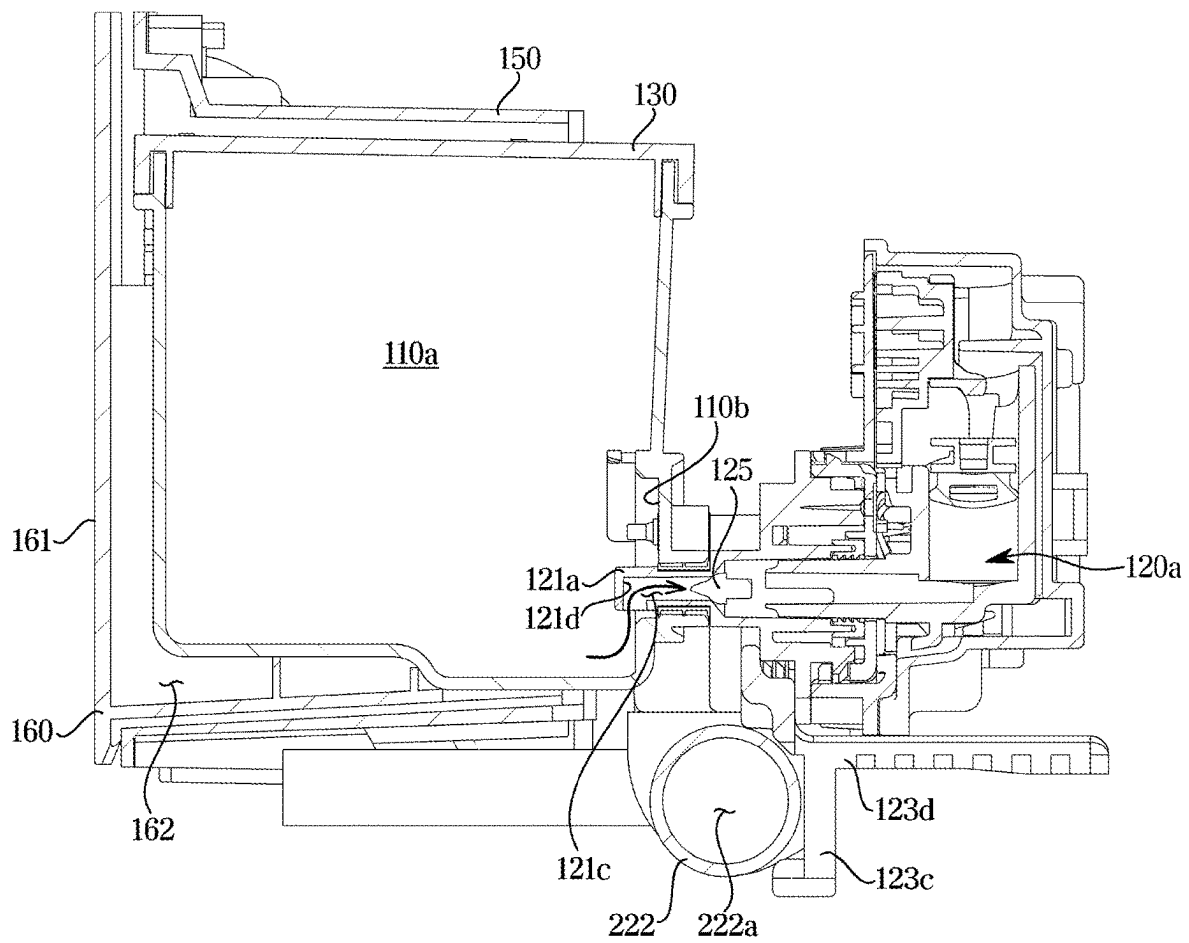


FIG. 12

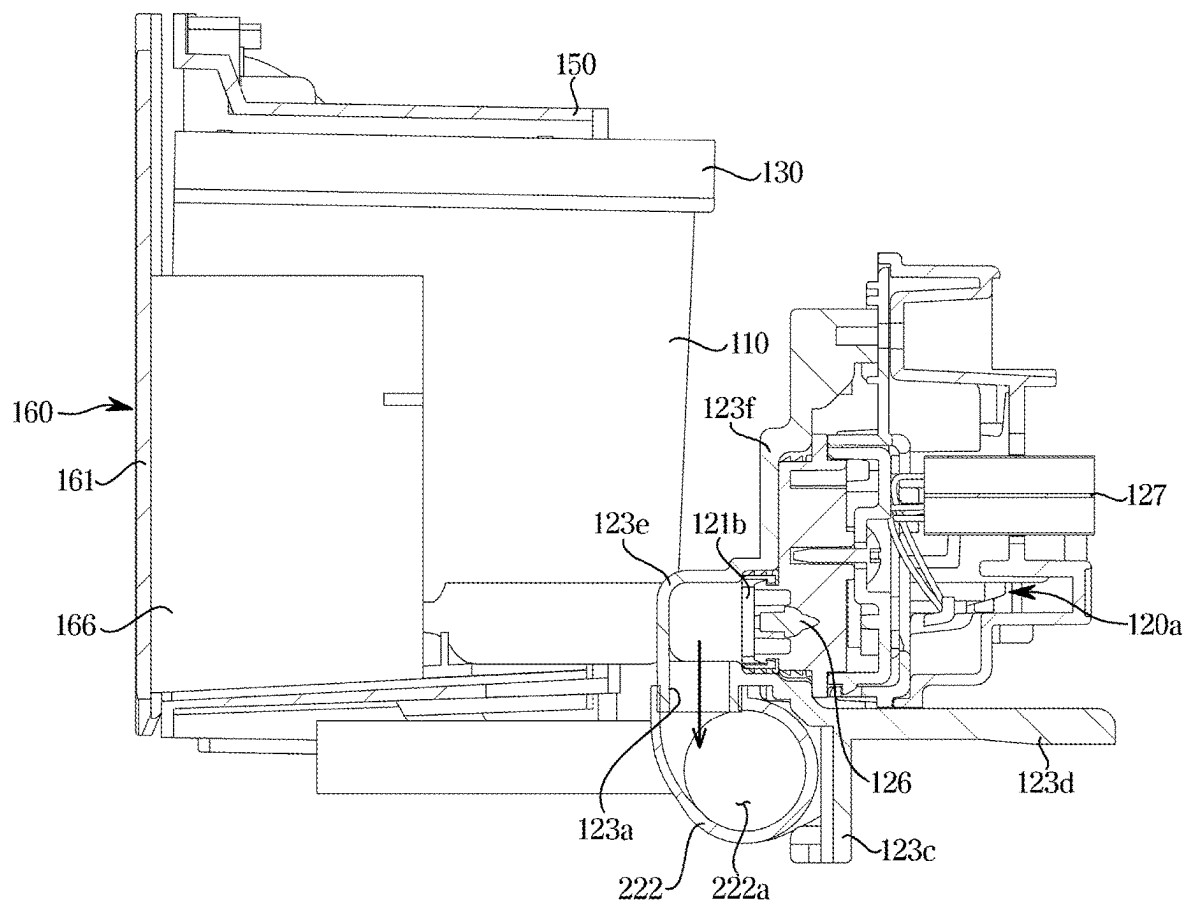


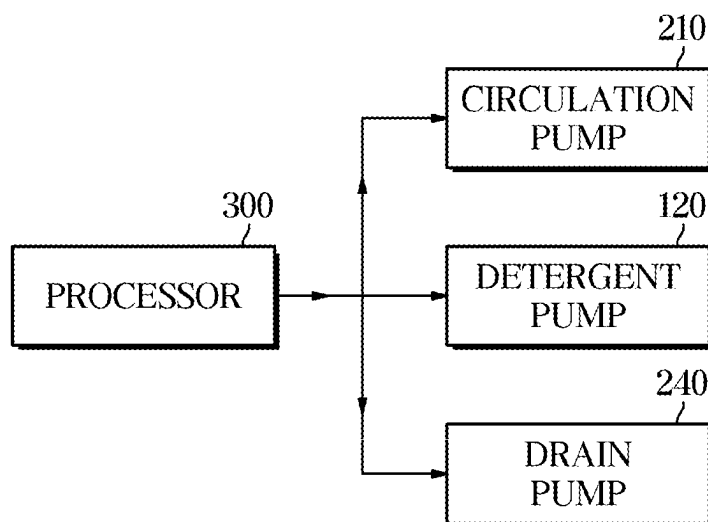
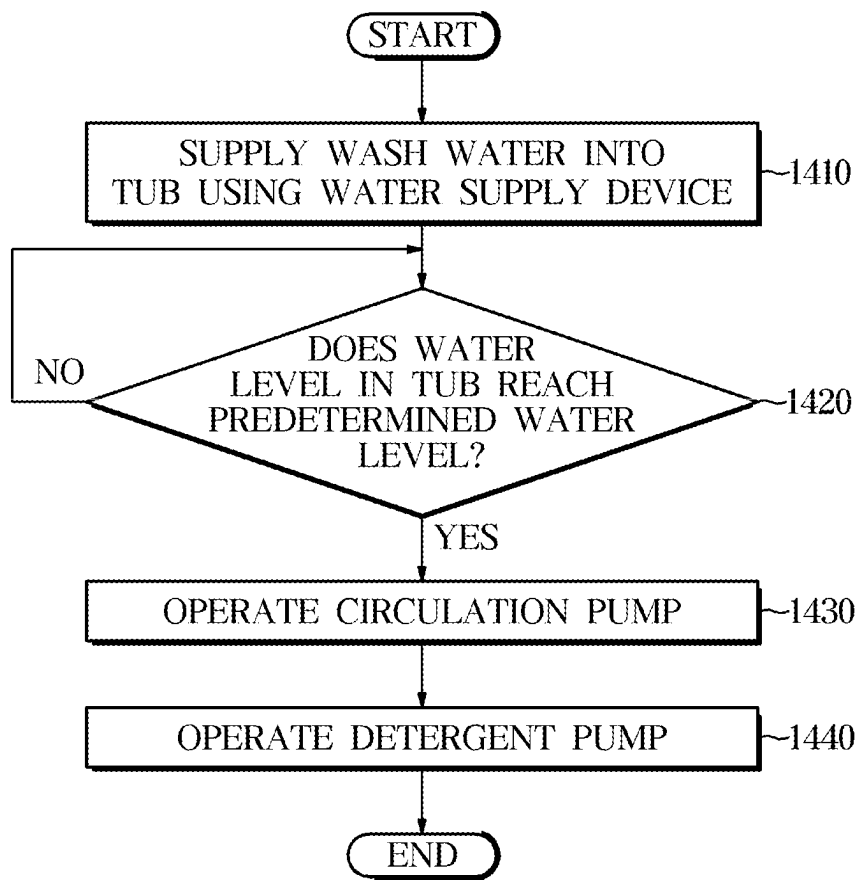
FIG. 13

FIG. 14

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CLOTHES TREATING APPARATUS**CROSS-REFERENCE TO RELATED APPLICATION**

This application is a continuation application, filed under 35 U.S.C. § 111(a), of International Application PCT/KR2022/014577 filed Sep. 28, 2022, and is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0000244, filed on Jan. 3, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND**1. Field**

The present disclosure relates to a clothes treating apparatus, and more particularly, to a clothes treating apparatus including a detergent supply device.

2. Description of the Related Art

In general, a clothes treating apparatus includes a washing machine configured to wash laundry loaded in a rotary tub by mutual friction by rotating the laundry and a detergent together using a driving force of a motor, a dryer configured to dry an object to be dried and contained in the rotary tub by rotating the rotary tub, and the like.

A clothes treating apparatus may include a detergent supply device provided in a cabinet, and the detergent supply device is a device configured to supply a detergent into a tub. The detergent supply device may include a detergent case, and a detergent pump configured to supply the detergent contained in the detergent case into the tub in response to docking with the detergent case.

In this case, the detergent discharged from the detergent pump should be mixed with wash water and the detergent may remain in a flow path from the detergent pump to a space in which the detergent is to be mixed with wash water. In the case where the detergent residues are left, contamination and solidification of the detergent may occur.

SUMMARY

In accordance with an aspect of the present disclosure, a clothes treating apparatus includes: a cabinet; a tub provided in the cabinet and configured to collect wash water; a circulation device configured to circulate the wash water contained in the tub, and the circulation device including a circulation pump, a first circulation path provided to guide wash water from the tub to the circulation pump, and a second circulation path provided to guide wash water from the circulation pump to the tub so that the wash water circulates by flowing from the tub to the circulation pump through the first circulation path and flowing back to the tub from the circulation pump through the second circulation path; and a detergent supply device configured to supply a detergent to the circulation device, and the detergent supply device including a detergent case, and a detergent pump connected to the detergent case to store the detergent and mounted on the first circulation path to supply the stored detergent through the first circulation path.

The first circulation path may include: a drain path to guide the wash water discharged from the tub out of the tub;

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and a mixing path to guide the wash water to the circulation pump flowing from the drain path and mounted with the detergent pump.

The drain path may include: a first drain path to guide the wash water from the tub out of the tub; and a second drain path connected to the first drain path and branching from the first drain path.

The clothes treating may further include a drain pump configured to receive the wash water from the first drain path and mounted on the second drain path.

The clothes treating apparatus may further include a water supply device configured to supply the wash water into the tub from an external water supply source, wherein the wash water flowing into the tub by the water supply device flows to the tub after passing through the first circulation path, the circulation pump and the second circulation path in response to the circulating of the wash water by the circulation pump.

The clothes treating apparatus may further include a processor configured to selectively operate the circulation pump or the drain pump.

The processor may be further configured to operate the circulation pump in response to determining that a water level in the tub reaches a predetermined water level.

The processor may be further configured to operate the detergent pump in response to the operating of the circulation pump.

The detergent case may include: a first detergent case; and a second detergent case separated from the first detergent case in a lateral direction, wherein the processor is further configured to operate the circulation pump based on initiation of a washing mode, the detergent pump to allow a detergent contained in the first detergent case to flow into the detergent pump based on the operation of the circulation pump, and the detergent pump to allow a detergent contained in the second detergent case to flow into the detergent pump based on initiation of a rinsing mode.

The processor may be further configured to turn off the circulation pump and turn on the drain pump to drain the wash water contained in the tub.

The detergent supply device and the circulation device may be disposed below the tub.

The detergent supply device may be a first detergent supply device, and the clothes treating apparatus may further include a second detergent supply device disposed above the tub to supply the detergent into the tub.

The clothes treating apparatus may further include: a fan above the tub to form a circulating air flow passing through the tub; and a heat exchanger above the tub to dehumidify and heat air discharged from the tub and supply the dehumidified and heated air to the tub.

The detergent supply device may further include a drawer configured to accommodate the detergent case and to be withdrawn from or inserted into the cabinet.

The drawer may include: a first drawer configured to accommodate the detergent case and withdrawn from or inserted into the cabinet; and a second drawer coupled to a side wall of the cabinet, configured to accommodate the first drawer, and slidably coupled to the first drawer allowing the first drawer to slide.

A clothes treating apparatus according to an embodiment of the present disclosure may include a cabinet, a tub in the cabinet, a water supply device configured to supply wash water into the tub from an external water source, and a detergent supply device configured to supply a detergent into the tub, disposed at a lower portion of the cabinet, the detergent supply device comprising a detergent case, a detergent pump including an inlet through which the deter-

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gent is introduced from the detergent case and an outlet through which the introduced detergent is to discharge the detergent, and a mixing space formed therein to receive wash water from the tub and directly connected to the outlet of the detergent pump to receive the discharged detergent from the detergent pump.

The supplied wash water from the external water supply source may flow back to the tub after passing through the water supply device, the tub, and the mixing space.

A clothes treating apparatus according to an embodiment of the present disclosure may include a cabinet, a tub in the cabinet and configured to collect wash water, a detergent supply device configured to supply a detergent into the tub, the detergent supply device including a detergent case to accommodate the detergent, a detergent pump including an inlet through which the detergent is introduced from the detergent case and an outlet through which the introduced detergent is discharged to discharge the detergent, and a mixing space formed therein to receive wash water from the tub and directly connected to the outlet of the detergent pump to receive the detergent from the detergent pump, a drain pump connected to the tub to drain the wash water contained in the tub out of the cabinet, a circulation pump connected to the tub to circulate the wash water, and a processor configured to selectively operate the drain pump or the circulation pump.

The processor may be further configured to operate the detergent pump in response to the operating of the circulation pump.

The processor may be further configured to turn off the circulation pump and turn on the drain pump to drain the wash water contained in the tub.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a clothes treating apparatus according to an embodiment of the present disclosure.

FIG. 2 is a side cross-sectional view of the clothes treating apparatus illustrated in FIG. 1.

FIG. 3 is a side cross-sectional view of the clothes treating apparatus illustrated in FIG. 1.

FIG. 4 is a perspective view showing a state where a detergent supply device is withdrawn from a cabinet in the clothes treating apparatus illustrated in FIG. 1.

FIG. 5 is a perspective view showing a state where the cabinet is omitted from the clothes treating apparatus illustrated in FIG. 1.

FIG. 6 is a perspective view of the clothes treating apparatus illustrated in FIG. 5.

FIG. 7 is a block diagram schematically illustrating the clothes treating apparatus illustrated in FIG. 1.

FIG. 8 is a perspective view of the detergent supply device and the circulation device of the clothes treating apparatus illustrated in FIG. 1.

FIG. 9 is an exploded perspective view of the detergent supply device and the circulation device illustrated in FIG. 8.

FIG. 10 is an exploded perspective view of the detergent pump illustrated in FIG. 9.

FIG. 11 is a cross-sectional view of the detergent supply device illustrated in FIG. 8.

FIG. 12 is a cross-sectional view of the detergent supply device illustrated in FIG. 8.

FIG. 13 is a control block diagram of the clothes treating apparatus illustrated in FIG. 1.

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FIG. 14 is a control flow chart of the clothes treating apparatus illustrated in FIG. 1.

DETAILED DESCRIPTION

Reference will now be made in detail to the embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. The embodiments described in the specification and shown in the drawings are only illustrative and are not intended to represent all aspects of the invention, such that various modifications may be made without departing from the spirit of the invention.

In the drawings, like reference numerals denote like elements or components having substantially same functions.

Throughout the specification, the terms used are merely used to describe particular embodiments, and are not intended to limit the present disclosure. As used herein, the singular forms are intended to include the plural forms as well, unless otherwise stated. Also, it is to be understood that the terms such as “include”, “have”, or the like, are intended to indicate the existence of the features, numbers, operations, components, parts, or combinations thereof disclosed in the specification, and are not intended to preclude the possibility that one or more other features, numbers, operations, components, parts, or combinations thereof may exist or may be added.

It will be understood that, although the terms “first”, “second”, etc., may be used herein to describe various elements, these elements should not be limited by these terms. The above terms are used only to distinguish one component from another. For example, a first component discussed below could be termed a second component, and similarly, the second component may be termed the first component without departing from the teachings of this disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

The terms used throughout the specification “front”, “rear”, “left”, and “right”, and the like are defined based on the drawings and the shape and position of each element are not limited by these terms. A clothes treating apparatus 1 according to various embodiments of the present disclosure may include washing machines, clothes dryers, clothes care apparatuses, and the like. However, for descriptive convenience, the embodiments of the present disclosure will be described below using a front-loading washing machine 1 as an example.

In addition, a detergent may include not only laundry detergents but also rinses, deodorizers, disinfectants, or fragrances for dryers.

A clothes treating apparatus may be capable of preventing detergent residues in a detergent supply device.

A clothes treating apparatus may be capable of reducing contamination and solidification of a detergent by preventing detergent residues.

A clothes treating apparatus may have improved user convenience.

According to an embodiment of the present disclosure, a clothes treating apparatus capable of preventing detergent residues by mounting a detergent pump on a circulation path may be provided.

According to an embodiment of the present disclosure, a clothes treating apparatus in which a detergent pump is directly connected to a circulation path may be provided.

According to an embodiment of the present disclosure, a clothes treating apparatus having improved user convenience

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nience by preventing malfunctioning caused by detergent residues in advance may be provided.

Hereinafter, the embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

FIG. 1 is a perspective view of a clothes treating apparatus according to an embodiment of the present disclosure. FIG. 2 is a side cross-sectional view of the clothes treating apparatus illustrated in FIG. 1. FIG. 3 is a side cross-sectional view of the clothes treating apparatus illustrated in FIG. 1. FIG. 4 is a perspective view showing a state where a detergent supply device is withdrawn from a cabinet in the clothes treating apparatus illustrated in FIG. 1.

Referring to FIGS. 1 to 4, a washing machine 1 includes a cabinet 10 defining an exterior appearance of the washing machine 1 and accommodating various components, a tub 20 provided in the cabinet 10 to contain mixed water in which wash water and a detergent are mixed, a drum 30 provided in the tub 20 to contain laundry and configured to rotate, and a drive device 60 configured to rotate the drum 30. The drive device 60 is located behind the tub 20 and configured to rotate the drum 30 in the forward-reverse directions to perform washing, rinsing, spin-drying, and drying operations.

The cabinet 10 may be formed in a substantially hexahedral shape. The cabinet 10 may be formed in a box shape with an open front. The cabinet 10 may include a rear plate 12d and both side plates 12b and 12c located at both sides of the rear plate 12d in the forward direction. Although the cabinet 10 including the rear plate 12d and the side plates 12b and 12c which are integrated with each other is shown herein by way of example, the present disclosure is not limited thereto. The cabinet 10 may include a front panel 11 provided in the open front, a base 19 provided at the bottom, and a top cover 12e provided at the upper surface.

The front panel 11 may include a first front panel 11a and a second front panel 11b located at an upper side of the first front panel 11a.

The first front panel 11a may be provided in front of the drum 30. The first front panel 11a may be located at a lower side of the second front panel 11b. The first front panel 11a may have an opening 13 through which laundry is loaded into the drum 30. The opening 13 of the first front panel 11a is formed to allow the laundry to be loaded or unloaded in the forward direction of the cabinet 10. The opening 13 of the first front panel 11a may be open or closed by a door 40. The door 40 may include a door frame 41 and a door glass 42. The door glass 42 may be formed of a transparent tempered glass material such that the interior of the cabinet 10 is visible therethrough.

The second front panel 11b may be provided at an upper-front side of the cabinet 10. The second front panel 11b may be disposed on the upper side of the first front panel 11a. The upper end of the second front panel 11b is provided to be coupled to the top cover 12e.

The second front panel 11b may include a control panel (not shown) including an inputter configured to receive an operation command from a user and a display (not shown) configured to display operation information of the washing machine 1.

The base 19 may be configured to cover the bottom of the cabinet 10.

A water supply device 14 may be provided above the tub 20. The water supply device 14 may include a water supply valve 14b configured to control water supply and water supply pipes 14a. The water supply device 14 may supply

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water into the tub 20. For example, water flowing into the cabinet 10 through the water supply device 14 may be collected in the cabinet 10.

The clothes treating apparatus according to an embodiment of the present disclosure may include detergent supply devices 90 and 100. The detergent supply devices 90 and 100 may supply detergents into the tub 20. The detergent supply devices 90 and 100 may be provided in plural. The plurality of detergent supply devices 90 and 100 may include a first detergent supply device 100 and a second detergent supply device 90.

The first detergent supply device 100 may be disposed below the tub 20. For example, the first detergent supply device 100 may be disposed at a lower portion in the cabinet 10. The first detergent supply device 100 may be disposed at the bottom in the cabinet 10. The first detergent supply device 100 may be withdrawn from the cabinet 10 or inserted into the cabinet 10 at a lower portion of the front panel. The first detergent supply device 100 may be slidably moved (See FIG. 4).

The first detergent supply device 100 may allow the detergent to flow into the tub 20. For example, the first detergent supply device 100 may allow a liquid laundry detergent and/or a fabric softener to flow into the tub 20. However, types of the detergent are not limited thereto.

The second detergent supply device 90 may be disposed above the tub 20. For example, the second detergent supply device 90 may be disposed at an upper portion in the cabinet 10. The second detergent supply device 90 may be disposed at the top in the cabinet 10. The second detergent supply device 90 may be installed in the second front panel 11b. The second detergent supply device 90 may be withdrawn from the cabinet 10 or inserted into the cabinet 10 at an upper portion of the front panel 11. The second detergent supply device 90 may be slidably moved.

The second detergent supply device 90 may allow the detergent to flow into the tub 20. For example, the second detergent supply device 90 may allow a solid laundry detergent and/or fabric softener to flow into the tub 20. However, types of the detergent are not limited thereto.

The first detergent supply device 100 may be disposed below the tub 20 and supply the detergent into the tub 20 during a water supplying process, and the second detergent supply device 90 may be disposed above the tub 20 and supply the detergent into the tub 20 during the water supplying process.

In the clothes treating apparatus according to an embodiment of the present disclosure 1, water flowing into the washing machine 1 through the water supply device 14 may flow into the detergent supply devices 90 and 100. For example, the second detergent supply device 90 may be connected to the tub 20 via a supply pipe 17. Wash water supplied via the water supply pipe 14a may pass through the second detergent supply device 90 and be mixed with the detergent, and the mixed water in which the wash water is mixed with the detergent may be supplied into the tub 20.

The washing machine may further include a circulation device 200. The circulation device 200 may allow water contained in the tub 20 to circulate inside the washing machine 1. For example, the circulation device 200 may allow water to flow from the tub 20 into the first detergent supply device 100 and then to flow from the first detergent supply device 100 back into the tub 20.

The drive device 60 may be disposed behind the drum 30. The drive device 60, as a component to rotate the drum 30, may be configured to rotate the drum 30 by transmitting a driving force generated by a motor 61 to a rotary shaft 67.

The motor **61** including a fixed stator **63** and a rotor **65** configured to rotate via electromagnetic interaction with the stator **63** may convert electric power into mechanical rotational power. However, the components of the motor **61** are not limited thereto.

The tub **20** stores the mixed water in which wash water is mixed with the detergent and may have a substantially cylindrical shape. The tub **20** may be fixed inside the cabinet **10**. The opening **13** of the front panel **11** may be connected to the tub **20** by a diaphragm **50**. The diaphragm **50** may seal a space between the front panel **11** and the tub **20**.

A spray nozzle **70** connected to a circulation hose and configured to spray the circulating mixed water into the tub **20** and/or the drum **30** may be provided at one side of the diaphragm **50**. The spray nozzle **70** may be provided to uniformly spray the mixed water into the entire laundry contained in the drum **30**. The circulation hose may be connected to a circulation pump **210** configured to pump the contained mixed water from the bottom of the tub **20**.

The drum **30** may perform a washing operation by lifting and dropping laundry while rotating in the tub **20**. A plurality of lifters **31** may be provided inside the drum **30**.

A drain device (not shown) configured to drain water contained in the tub **20** and including a drain pipe **241** (See FIG. 6) and a drain valve (not shown) may be installed below the tub **20**.

The tub **20** may be provided to be elastically supported by the cabinet **10** via a spring (not shown) located at an upper portion of the tub **20** and dampers **80** provided at lower portions of the tub **20**. That is, the spring and the dampers **80** absorb vibration energy between the tub **20** and the cabinet **10** while vibration generated during rotation of the drum **30** is transmitted to the tub **20** and the cabinet **10**, thereby attenuating the vibration transmitted to the tub **20** and the cabinet **10**.

The washing machine **1** may include a hot air supply device **170**. The hot air supply device **170** may be disposed in the cabinet **10**. The hot air supply device **140** may supply hot air to the tub **20** and/or the drum **30**. The hot air supply device **170** may include a fan **172** and a heat exchanger **171**.

The washing machine **1** according to an embodiment of the present disclosure may include a compressor (not shown), a heat exchanger **171**, and an expansion device (not shown). The heat exchanger **171** may be located above the drum **30**. The heat exchanger **171** may include a condenser (not shown) and an evaporator (not shown).

The washing machine may further include the fan **172**. The fan **172** may blow air inside the drum interior **30a** and air passing through the heat exchanger **171** in the air circulation path **190**. The fan **172** may form an air flow such that air flows in the air circulation path **190**. The fan **172** may be a centrifugal fan. However, types of the fan **172** are not limited thereto.

The heat exchanger **171** may be configured to dehumidify and heat air passing through the air circulation path **190**. The heat exchanger **171** may dehumidify and heat air flowing into a tub interior **20c**. Thereby, clothes contained in the drum interior **30a** may be dried.

The washing machine may include a duct assembly **180**. The duct assembly **180** may further include an upper duct **181**, a front duct **182**, and a rear duct **183**.

The upper duct **181** may be located above the drum **30** and the tub **20** and guide air discharged from the drum **30** to flow back into the tub **20** after dehumidifying and heating the air. The heat exchanger **171** may be accommodated in the upper

duct **181**. The upper duct **181** may be referred to as an upper frame **181**. A first air circulation path **191** may be provided in the upper duct **181**.

The front duct **182** may be located in front of the drum **30** and the tub **20** and guide air flowing toward the drum **30**. Air that has passed the heat exchanger **171** may be supplied into the drum **30** through the front duct **182**. The front duct **182** may constitute a part of the air circulation path **190**. The fan **172** may be accommodated in the front duct **182**. A second air circulation path **192** may be provided in the front duct **182**.

The rear duct **183** may be located behind the drum **30** and the tub **20** and guide air from the drum interior **30a** to the heat exchanger **171**. Air contained in the drum interior **30a** may flow into the heat exchanger **171** through the rear duct **183**. The rear duct **183** may constitute a part of the air circulation path **190**. A third air circulation path **193** may be provided in the rear duct **183**.

The front duct **182** and the rear duct **183** may guide air contained in the drum interior **30a** to circulate through the air circulation path **190** in the cabinet **10**. In addition, because air discharged from the front duct **182** may be introduced into the tub **20** and the drum **30** through the diaphragm **50**, the diaphragm **50** may also circulate air inside the cabinet **10** through the air circulation path **190**.

The washing machine may further include the air circulation path **190**. The air circulation path **190** may include a first air circulation path **191**, a second air circulation path **192**, and a third air circulation path **193**. The first air circulation path **191** may be formed by the upper duct **181**, the second air circulation path **192** may be formed by the front duct **182**, and the third air circulation path **193** may be formed by the rear duct **183**. The first air circulation path **191**, the second air circulation path **192**, and the third air circulation path **193** are not limited to the above-described example. For example, a flow path formed in the rear duct **183** may be the first air circulation path **191**, a flow path formed in the upper duct **181** may be the second air circulation path **192**, and a flow path formed in the front duct **182** may be formed in the third air circulation path **193**.

Hereinafter, an air flow in the washing machine while the drying mode of the washing machine is performed will be described. After the washing and spin-drying modes of the washing machine are finished, a drying mode may be performed. A user may add the drying mode before the washing mode of the washing machine is initiated.

Air in the drum interior **30a** may circulate in a closed circuit (in a direction of the arrow) inside the cabinet **10**. In response to the operation of the fan **172**, humid air in the drum interior **30a** may flow into the tub **20** through spin-drying holes **32** formed in the circumferential inner wall of the drum **30** and/or through-holes **33** formed in a drum rear wall **30b**. For example, air in the drum interior **30a** may flow into a space **20a** between the drum **30** and the tub **20**. Air in the space **20a** between the drum **30** and the tub **20** may flow into the air circulation path **190** through communicating holes **183c** formed in a tub rear wall **20b**. Air may pass through the third air circulation path **193** located behind the tub **20** and the drum **30**. After air flows to the top end of the third air circulation path **193**, the air may flow in the first air circulation path **191**. The air may sequentially pass through the heat exchanger **171** in the first air circulation path **191**. Air having a high humidity by the clothes of the drum interior **30a** may become hot air having a low humidity while passing through the heat exchanger **171**. The hot air with a low humidity may be blown by the fan **172** disposed at a downstream area of the heat exchanger **171**. Air that has

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passed through the fan 172 may flow in the second air circulation path 192 in front of the tub 20 and the drum 30. Air in the second air circulation path 192 may pass through a communicating hole 182c formed at the diaphragm 50 at the bottom end of the second air circulation path 192 and flow back into the drum 30 after passing through the diaphragm 50. The hot air with a low humidity may become humid air again by the clothes contained in the drum interior 30a and then turn to hot air with a low humidity again by the above-described process.

FIG. 5 is a perspective view showing a state where the cabinet is omitted from the clothes treating apparatus illustrated in FIG. 1. FIG. 6 is a perspective view of the clothes treating apparatus illustrated in FIG. 5. FIG. 7 is a block diagram schematically illustrating the clothes treating apparatus illustrated in FIG. 1.

Referring to FIGS. 5 to 7, the clothes treating apparatus according to an embodiment of the present disclosure may include a water supply device 14, a tub 20, a circulation device 200, and a detergent supply device 100. In the following description, the detergent supply device 100 may be the first detergent supply device 100.

The circulation device 200 may include a circulation pump 210, a first circulation path 220, and a second circulation path 230. The circulation device 200 may allow water contained in the tub interior 20c to flow out of the tub 20 and back to the tub interior 20c. For example, the circulation device 200 may be disposed below the tub 20 and circulate water contained in the tub interior 20c. However, the position of the circulation device 200 is not limited to the above-described example.

In addition, although a plurality of separate circulation paths 220 and 230 are illustrated in the drawings, the embodiment is not limited thereto, and the circulation paths 220 and 230 may be integrated with each other. For example, the first circulation path 220 or the second circulation path 230 may be omitted and wash water may circulate via another circulation path. The plurality of circulation paths 220 and 230 may be referred to as a plurality of circulation guides 220 and

The circulation pump 210 may be connected to the plurality of circulation paths 220 and 230 to allow water contained in the circulation paths 220 and 230 to flow into tub 20. The circulation pump 210 may be located at one side of the plurality of circulation paths 220 and 230. For example, the circulation pump 210 may be located on the left side (e.g., in the Y direction) of the first circulation path 220 under the second circulation path 230. The circulation pump 210 may operate optionally with a drain pump 240, which will be described below.

The first circulation path 220 may be provided to guide wash water from the tub 20 to the circulation pump 210. The first circulation path 220 may be disposed under the tub 20 and guide water drained from the tub interior 20c to the circulation pump 210. The first circulation path 220 may include a drain path 221 and a mixing path 222. The first circulation path 220 may be referred to as a first circulation guide 220.

The drain path 221 may be disposed under the tub 20 and guide water drained from the tub interior 20c. An inner flow path 224 may be formed in the drain path 221. The drain path 221 may guide water drained from the tub 20 to flow into the circulation pump 210 or to be discharged out of the washing machine 1. The drain path 221 may be referred to as a drain guide 221.

The drain path 221 may include a first drain path 221a and a second drain path 221b. The first drain path 221a may

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guide water discharged from the tub 20 to the second drain path 221b. The second drain path 221b may guide water discharged from the tub 20 to flow into the circulation pump 210 or to be discharged out of the washing machine. The inside of the second drain path 221b may be open or closed by the drain pump 240. The second drain path 221b may serve as a filter case 234 to which a filter is coupled. The drain pump 240 may be coupled to the second drain path 221b. The first drain path 221a may be referred to as a first drain guide 221a and the second drain path 221b may be referred to as a second drain guide 221b.

The mixing path 222 may be connected to the drain path 221. For example, the mixing path 222 may be connected to the second drain path 221b and guide wash water to the circulation pump 210. The mixing path 222 may be connected to the detergent supply device 100. For example, a detergent contained in the detergent case 110 of the detergent supply device 100 may be sucked into a detergent pump 120 and discharged into a mixing space 222a in the mixing path 222. Detailed descriptions thereof will be given below. Therefore, wash water and the detergent may be mixed in the mixing path 222, and the wash water mixed with the detergent may flow into the tub 20 after passing through the mixing path 222, the circulation pump 210, and the second circulation path 230. The mixing path 222 may be mounted on the detergent pump 120 which will be described below. Thus, the mixing path 222 may be directly connected to the detergent pump 120 without a separate part connecting the mixing path 222 with the detergent pump 120, and detergent residues may be minimized. The mixing path 222 may be referred to as a mixing guide 222.

The second circulation path 230 may be provided to guide wash water from the circulation pump 210 to the tub 20. The second circulation path 230 may be disposed under the tub 20 and guide wash water from the circulation device 200 to the tub interior 20c. The second circulation path 230 may be connected to the first circulation path 220 and the tub 20, respectively. The second circulation path 230 may be referred to as a second circulation guide 230. The second circulation path 230 may include a circulation pump case 231 and a supply pipe 232. Detailed descriptions thereof will be given below.

The detergent supply device 100 may be disposed at one side of the circulation device 200. The detergent supply device 100 may supply the detergent to the circulation device 200. The detergent supplied to the circulation device 200 may be mixed with wash water in the circulation device 200. The mixed detergent may flow into the tub 20 through the plurality of circulation paths 220 and 230.

The detergent supply device 100 may include the detergent pump 120 configured to suck a detergent from the detergent case 110 after docking with the detergent case 110, a housing 160 accommodating the detergent case 110 and provided to be withdrawn from the cabinet 10 and inserted into the cabinet 10, and a second drawer 150 accommodating the housing 160. The housing 160 may slide out from the inside of the second drawer 150. The second drawer 150 may be coupled to side walls 12b and 12c or side plates 12b and 12c of the cabinet 10.

The drain pump 240 may drain water discharged from the tub 20 out of the washing machine through the drain pipe 241. The drain pump 240 may be coupled to the second drain path 221b to open or close the inner flow path 224 in the second drain path 221b. The drain pump 240 may be turned on/off optionally with the circulation pump 210 by the detergent pump 120. For example, to circulate water in the washing machine, the circulation pump 210 may be driven.

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On the contrary, to drain water, only the drain pump **240** may be driven, and therefore water may be drained out of the washing machine.

The washing machine may further include a switch. The switch may sense whether a first drawer is accommodated in the cabinet **10**. For example, in response to insertion of that the second drawer **150** into the cabinet **10**, a button (not shown) of the switch may be in a pressed state. In response to withdrawing of the second drawer **150** from the cabinet **10**, the button (not shown) may be in a non-pressed state.

The washing machine may further include a sensor guide **225**. The sensor guide **225** may protrude from the first drain guide **221a**. The sensor guide **225** may be connected to a water level sensor **250**. For example, the sensor guide **225** may be connected to the water level sensor **250** via a separate hose. For example, the water level sensor **250** may be an air sensor, and the water level in the sensor guide **225** may increase as the water level in the tub **20c** increases. Accordingly, a pressure in the sensor guide **225** may increase in accordance with the increasing water level, and the water level sensor **250** may sense the water level of the tub interior **20c** by sensing the same.

Referring to FIGS. **5** to **7**, wash water circulating in the cabinet **10** will be described.

Water introduced from the outside of the washing machine **1** may enter the tub **20** through the water supply device **14**. For example, the water supply device **14** may be disposed above the tub **20** to allow water to flow into the interior **20c** of the tub **20**.

The water level sensor **250** provided at one side of the tub **20** may sense an amount of water collected in the tub interior **20c**. For example, the water level sensor **250** may be coupled to an upper portion of the tub **20** and sense the amount of wash water of the tub interior **20c**. For example, upon determination that the water level of the tub **20** sensed by the water level sensor **250** is $\frac{1}{3}$ or more, a processor **300** may operate the circulation pump **210** in response thereto. Alternatively, the processor **300** may operate the drain pump **240** to drain water contained in the tub interior **20c** out of the washing machine. In this case, the first circulation path **220**, the second circulation path **230**, and the like below the tub **20** may be filled with water in the cabinet **10**.

Although the water level sensor **250** disposed in the interior **20c** of the tub **20** is illustrated in the drawing, the position of the water level sensor **250** is not limited to the above-described example.

Hereinafter, the embodiment will be described based on the operation of the circulation pump **210**.

In response to the operation of the circulation pump **210**, water contained in the tub interior **20c** may flow to the circulation device **200** through the drain hole **223** disposed under the tub **20**. For example, water contained in the tub interior **20c** may flow to the inner flow path **224** through the first circulation path **220** or the drain hole **223** of the drain path **221**.

Wash water may be discharged from the tub **20** through the first drain path **221a** and guided to the second drain path **221b**. Because a filter (not shown) is provided in the second drain path **221b**, wash water may be filtered by the filter. Wash water guided to the second drain path **221b** may be guided to the mixing path **222**. In this regard, in response to the operation of the detergent pump **120**, the detergent may be discharged to mixing space **222a** in the mixing path **222** from the detergent pump **120**. Wash water may be mixed with the detergent while passing through the mixing path **222**.

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Wash water mixed with the detergent may be guided toward the circulation pump **210**. Wash water may flow to the second circulation path **230** through the circulation pump **210**. Because the second circulation path **230** may be connected to the first circulation path **220** and the tub **20**, respectively, wash water that has passed through the first circulation path **220** and the circulation pump **210** may flow into the tub interior **20c**.

The embodiment will be described based on the operation of the drain pump **240**.

In response to the operation of the drain pump **240**, water contained in the tub interior **20c** may flow to the circulation device **200** through the drain hole **223** disposed under the tub **20**. For example, water contained in the tub interior **20c** may flow to the inner flow path **224** through the first circulation path **220** or the drain hole **223** of the drain path **221**.

Wash water may be discharged from the tub **20** through the first drain path **221a** and guided to the second drain path **221b**. Because a filter (not shown) is provided in the second drain path **221b**, wash water may be filtered by the filter. The drain pump **240** may serve as a valve to open and close the inner flow path **224**, and the inner flow path **224** may be open and closed by operating the drain pump **240**. In response to the opening of the inner flow path **224**, wash water guided to the second drain path **221b** may be guided out of the washing machine **1** through the drain pipe **241**. In this case, the detergent pump **120** and/or the circulation pump **210** may be in an off-state.

FIG. **8** is a perspective view of the detergent supply device and the circulation device of the clothes treating apparatus illustrated in FIG. **1**. FIG. **9** is an exploded perspective view of the detergent supply device and the circulation device illustrated in FIG. **8**.

Referring to FIGS. **8** and **9**, the clothes treating apparatus according to an embodiment of the present disclosure may include the detergent supply device **100** and the circulation device **200**.

The detergent supply device **100** may include a detergent case **110**, a detergent pump **120**, a detergent case cover **130**, a cap **140**, a second drawer **150**, and a housing **160**.

The detergent case **110** may be docked with the detergent pump **120**. Upon docking of the detergent case **110** with the detergent pump **120**, the detergent of the accommodating space **110a** of the detergent case **110** may be discharged into the mixing path **222** via the detergent pump **120**.

The detergent case **110** may be inserted into the second drawer **150** or withdrawn from the second drawer **150** in a state of being accommodated in the housing **160**. The detergent case **110** may be inserted into a receiving space **151** or withdrawn from the receiving space **151** via the opening **150a** of the second drawer **150**. The detergent case **110** and the housing **160** may be slid out from the second drawer **150**.

The detergent case **110** may include a first detergent case **111**, a second detergent case **112**, and a pump connector **111b**.

The detergent case **110** may store a main detergent (powder detergent or liquid detergent), a preliminary detergent, a fabric softener and/or a bleach.

A main laundry detergent, a fabric softener, and/or a rinse may be contained in the first detergent case **111** and/or the second detergent case **112**. For example, the main laundry detergent may be contained in the first detergent case **111**, and the fabric softener or the preliminary detergent may be contained in the second detergent case **112**. However, types of the detergent contained therein and/or size of the detergent case **110** are not limited to the above-described

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example. The first detergent case **111** and the second detergent case **112** may be disposed inside the housing **160**. The first detergent case **111** may form a first accommodating space **111a** and the second detergent case **112** may form a second accommodating space **112a**. For example, the main laundry detergent may be contained in the first accommodating space **111a**, and the fabric softener, the preliminary detergent, or the like may be contained in the second accommodating space **112a**.

Although the drawings illustrate a plurality of detergent cases **110** partitioned and separated from each other, the embodiment is not limited thereto.

The pump connector **111b** may be formed at the rear of the first detergent case **111** and at the rear of the second detergent case **112**, respectively. The pump connector **111b** may serve as an outlet **111b**. Although the outlet **111b** is formed only at the first detergent case **111** in the drawing, the outlet **111b** may also be formed at the second detergent case **112**. The outlet **111b** may allow the detergent contained in the first accommodating space **111a** and/or the second accommodating space **112a** to flow into the detergent pump **120** via a pipe (not shown). For example, upon docking of the detergent case **110** with the detergent pump **120**, the detergent may flow into the detergent pump **120** through the outlet **111b** and then into the mixing space **222a** of the mixing path **222**. However, the position of the outlet **111b** is not limited to that shown in the drawing.

The detergent case cover **130** may be configured to cover the top of the detergent case **110**. For example, the detergent case cover **130** may be configured to cover the top end of the detergent case **110**. The detergent case cover **130** may include a first detergent case cover **131** configured to cover the top of the first detergent case **111** to correspond to the first accommodating space **111a**, and a second detergent case cover **132** configured to cover the top of the second detergent case **112** to correspond to the second accommodating space **112a**.

The cap **140** may be coupled to the detergent case cover **130**. The cap **140** may be removed in the case where the user adds a detergent into the interior **111a** and **112a** of the detergent case **110**. Upon completion of adding the detergent into the interior **111a** and **112a** of the detergent case **110** by the user, the cap **140** may be mounted on the detergent case cover **130** again. The interior **111a** and **112a** of the detergent case **110** may serve as the accommodating spaces **111a** and **112a** of the detergent case **110**.

The cap **140** may include a first cap **141** and a second cap **142**. The first cap **141** may be mounted on the first detergent case cover **131**, and the second cap **142** may be mounted on the second detergent case cover **132**.

The detergent pump **120** may be disposed behind the second drawer **150** and the detergent case **110**. Upon docking with the detergent case **110** inserted into the second drawer **150**, the detergent pump **120** may suck the detergent contained in the detergent case **110** and discharge the sucked detergent into the mixing space **222a** in the mixing path **222**. For example, upon docking of the outlet **111b** of the detergent case **110** with first mover **121a** of the detergent pump **120**, the detergent contained in the detergent case **110** may flow into the detergent pump **120** through the outlet **111b** and the first mover **121a** and then flow into the mixing space **222a** through the second mover **121b**. The detergent flowing into the mixing space **222a** may be mixed with water entering the tub interior **20c** through the mixing path **222**.

The second drawer **150** may accommodate the housing **160** and the detergent case **110**. The housing **160** and the detergent case **110** may be withdrawn from or inserted into

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the second drawer **150** via the opening **150a**. The second drawer **150** may serve as the case **150**. The second drawer **150** is coupled to a side wall of the cabinet **10** to accommodate the first drawer and may be slidably coupled to the first drawer such that the first drawer slides.

The second drawer **150** may include a receiving space **151**, a sliding coupler **152**, a filter coupler **153**, a rail coupler **154a**, and a stopper **155**.

The receiving space **151** may be formed inside the second drawer **150**. The housing **160** and the detergent case **110** may be disposed in the receiving space **151**.

The sliding couplers **152** may be formed from both side walls **154** of the second drawer **150**. The sliding couplers **152** may extend backward from both side walls **154**. The sliding couplers **152** may be coupled to side plates **12b** and **12c** of the cabinet **10**. The sliding coupler **152** may be provided in plural. A plurality of sliding couplers **152** may extend backward from the left wall **154** and/or the right wall **154** of the second drawer **150**.

The filter coupler **153** may be formed by opening a part of a rear wall **150b** of the second drawer **150**. The filter case **234** may be coupled to the filter coupler **153**.

The rail coupler **154a** may be formed by being recessed from an inner wall **154** of the second drawer **150**. The rail coupler **154a** may be provided in plural. For example, the rail coupler **154a** may be formed by being recessed from inner surfaces of the left wall **154** and/or the right wall **154** of the second drawer **150**. The rail coupler **154a** may be slidably coupled to a rail **164** of the housing **160**. For example, the rail **164** of the housing **160** may be coupled to the rail coupler **154a** and the housing **160** may slide.

The stopper **155** may fix the rail **164** in a state where the second drawer **160** is inserted into or accommodated in the cabinet **10**. The stopper **155** may be mounted at the sliding coupler **152**. The stopper **155** may be provided in plural.

The housing **160** may accommodate the detergent case **110**, the detergent case cover **130**, and the cap **140**. The housing **160** may be slidably coupled to the second drawer **150**. The user may add the detergent into the interior **111a** and **112a** of the detergent case **110** accommodated in the housing **160** by pulling the housing **160** from the cabinet **10** and then insert the housing **160** into the cabinet **10** again.

The housing **160** may serve as a first drawer. The housing **160** may be provided to be withdrawn from or inserted into the cabinet **10**.

The housing **160** may be provided in plural. The plurality of housings **160** may include a first housing **161** and a second housing **162**. The first housing **161** may accommodate the first detergent case **111**, and the second housing **162** may accommodate the second detergent case **112**. Each of the housings **161** and **162** may include a handle **165a** or **165b**, an accommodating space **161c** and **162c**, and a rail **164**.

The handles **165a** and **165b** may be formed at the front surfaces **161a** and **161b** of the housing **160**. The handles **165a** and **165b** may be recessed from the front surfaces **161a** and **161b**. The handles **165a** and **165b** may be provided in plural. The first handle **165a** may be formed at the front surface **161a** of the first housing **161**, and the second handle **165b** may be formed at the front surface **161b** of the second housing **162**. The user may hold the detergent case **110** via the handles **165a** and **165b** and insert the detergent case **110** into the cabinet **10** or withdraw the detergent case **110** from the cabinet **10**.

The second housing **162** may accommodate the detergent case **110**, the detergent case cover **130**, and the cap **140**. The top end of the second housing **162** may be open. The

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detergent case **110**, the detergent case cover **130**, and the cap **140** may be separated from the housing **160** through the top side of the accommodating space **162**.

The rail **164** may be coupled to a side wall **163** of the housing **160**. The rail **164** may be provided in plural. For example, the first rail **164** may be coupled to a left wall **163** of the first housing **161**, and the second rail **164** may be coupled to a right wall of the second housing **162**.

The circulation device **200** may circulate water contained in the tub interior **20c**. For example, the circulation device **200** may allow water contained in the tub interior **20c** to flow out of the tub **20** and back into the tub interior **20c**. The circulation device **200** may include the circulation pump **210**, the first circulation path **220**, and the second circulation path **230**.

The circulation pump **210** may be disposed on the left below the tub **20**. The circulation pump **210** may pump wash water to introduce wash water flowing in the circulation device **200** into the tub **20**. The circulation pump **210** may be mounted on the circulation pump case **231**. For example, the circulation pump **210** may be mounted on a circulation pump mount **231a** provided at the circulation pump case **231**. The circulation pump **210** may open or close flow paths **222a** and **232a** provided in the circulation device **200**.

The first circulation path **220** may allow wash water to flow into the circulation pump **210** from the tub **20**. The first circulation path **220** may include the drain path **221** and the mixing path **222**.

The drain path **221** may allow water discharged from the tub **20** to flow into the circulation pump **210** or to flow out of the washing machine. The drain path **221** may include the first drain path **221a** and the second drain path **221b**.

The first drain path **221a** may guide water discharged from the tub **20** to the second drain path **221b**. A first end of the first drain path **221a** may be connected to the tub **20**, and a second end **226** of the first drain path **221a** may be connected to the second drain path **221b**. The drain hole **223** may be formed at the first end of the first drain path **221a**. The inner flow path **224** may be formed in the first drain path **221a**.

The second drain path **221b** may guide water discharged from the tub **20** to flow into the circulation pump **210** or to flow out of the washing machine. The second drain path **221b** may be connected to the first drain path **221a**. The second drain path **221b** may branch the inner flow path **224** formed in the first drain path **221a**. The second drain path **221b** may serve as a filter case **234** to which a filter is coupled.

The filter case **234** may include a filter mount **234a**, a drain pump mount **234b**, a mixing guide mount **234c**, a first drain guide connector **234d**, and a spray nozzle connector **234e**. Also, the drain pipe **241** (See FIG. 6) may be formed at one side of the filter case **234**. The drain pipe **241** may be a first drain pipe **241**.

The filter mount **234a** may be formed in the filter case **234**. A filter (not shown) may be mounted on the filter mount **234a** to filter water discharged from the tub **20**.

The drain pump **240** may be mounted on the drain pump mount **234b**. The drain pump **240** may open or close the inner flow path **224**. For example, the drain pump **240** may open or close the flow path in the filter case **234**. In response to the operation of the drain pump **240**, water that has passed through the first drain path **221a** and the second drain path **221b** may flow out of the washing machine. The drain pump mount **234b** may be formed on the right side of the filter case **234**.

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The mixing path **222** may be mounted on the mixing guide mount **234c**. For example, the mixing guide mount **234c** may be connected to a second drain guide connector **222b** of the mixing path **222**. Accordingly, water that has passed through the drain path **221** may flow into the mixing path **222**. The mixing guide mount **234c** may be formed on the left side of the filter case **234**.

The first drain guide connector **234d** may be connected to the first drain path **221a**. For example, the first drain guide connector **234d** may be connected to the second end **226** of the first drain path **221a**.

The spray nozzle connector **234e** may be connected to the spray nozzle **70** spraying wash water into the tub **20** and/or the drum **30**. The spray nozzle connector **234e** may protrude upward.

The mixing path **222** may guide water that has passed through the drain path **221** to the circulation pump **210**. The mixing space **222a** may be formed in the mixing path **222**. In the mixing space **222a**, wash water may be mixed with the detergent introduced into the mixing space **222a** from the detergent case **110** through the detergent pump **120**. The mixing path **222** may be disposed under the detergent case **110** and the detergent pump **120** (See FIGS. 11 and 12). The mixing path **222** may be directly connected to the detergent pump **120** without using a hose or pipe interposed therebetween. Therefore, the possibility of solidification of a detergent remaining in a hose or pipe, which connects the mixing path **222** with the detergent pump **120**, may be significantly reduced.

The mixing path **222** may include a mixing space **222a**, a second drain guide connector **222b**, a circulation pump communicator **222c**, and a drain pipe **222e**. The mixing space **222a** may be formed in the mixing path **222** and wash water may be mixed with the detergent therein.

The second drain guide connector **222b** may be disposed on the right side of the mixing path **222** and connected to the mixing guide mount **234c**. Thereby, wash water contained in the drain path **221** may flow into the mixing space **222a** and the detergent may be mixed with wash water.

The circulation pump communicator **222c** may be disposed on the left side of the mixing path **222** to be connected to the circulation pump **210** and/or the circulation pump case **231**. Thereby, wash water contained in the mixing space **222a** may be pumped by the circulation pump **210** to flow into the tub **20**.

The drain pipe **222e** may be a second drain pipe **222e**. The second drain pipe **222e** may be provided to remove water remaining after the rinsing operation of the washing machine. For example, the second drain pipe **222e** may be blocked with a plug, and the user may remove water remaining in the mixing space **222a** by removing the plug, if required.

The second circulation path **230** may be provided to guide wash water from the circulation pump **210** to the tub **20**. The second circulation path **230** may be connected to the first circulation path **220** and the tub **20**, respectively. The second circulation path **230** may include the circulation pump case **231** and the supply pipe **232**.

The circulation pump case **231** may accommodate a part of the circulation pump **210**. The circulation pump case **231** may include a circulation pump mount **231a** and a supply pipe mount **231b**.

The circulation pump mount **231a** may be formed at one side of the circulation pump case **231** in an open state. For example, the circulation pump mount **231a** may be formed

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on the left side of the circulation pump case 231. The circulation pump 210 may be mounted on the circulation pump mount 231a.

The supply pipe mount 231b may extend upward from the circulation pump case 231. The supply pipe mount 231b may protrude toward the supply pipe 232. For example, the supply pipe mount 231b may be connected to a first end 232b of the supply pipe 232. The first end 232b of the supply pipe 232 may be a lower end 232b. The lower end 232b of the supply pipe 232 may serve as a circulation pump connector 232b.

The supply pipe 232 may allow wash water pumped by the circulation pump 210 to flow into the tub 20 from the circulation pump case 231. A supply flow path 232a may be formed in the supply pipe 232. The supply flow path 232a may connect the mixing space 222a with the interior 20c of the tub 20. The supply pipe 232 may be connected to the circulation pump 210 and the tub 20, respectively. The first end 232b of the supply pipe 232 may serve as a circulation pump connector 232b connected to the circulation pump 210. A second end 232c of the supply pipe 232 may serve as a tub connector 232c connected to the tub 20. The second end 232c of the supply pipe 232 may be an upper end 232c.

FIG. 10 is an exploded perspective view of the detergent pump illustrated in FIG. 9.

Referring to FIG. 10, the detergent supply device 100 may include the detergent pump 120 in the washing machine according to an embodiment of the present disclosure.

The detergent pump 120 may include a pump 120a and a mounting member 123. The mounting member 123 may be mounted on the front of the pump 120a. The mounting member 123 may be a cover member 123 covering the pump 120a.

A mover 121 may be formed at the front of the pump 120a. The mover 121 may include a first mover 121a and a second mover 121b. The first mover 121a may allow the detergent to flow into the pump 120a from the detergent case 110. The second mover 121b may allow the detergent to flow into the mixing space 222a from the pump 120a. The first mover 121a and the second mover 121b may protrude forward. However, the shapes of the first mover 121a and the second mover 121b are not limited to the above-described example.

The mounting member 123 may support the pump 120a. In addition, the mounting member 123 may cover the front of the pump 120a to protect the second mover 121b and allow the second mover 121b to communicate with the mixing path 222. The mounting member 123 may be mounted on the mixing path 222.

The mounting member 123 may include supports 123c and 123d and a mount. The supports 123c and 123d may be provided in plural. The plurality of supports 123c and 123d may include a first support 123c and a second support 123d.

The first support 123c may support the pump 120a at the bottom plate 19 and/or bottom. The first support 123c may vertically extend. The second support 123d may be bent from the top end of the first support 123c and extend backward. The second support 123d may support the bottom of the pump 120a.

The mounts 123e and 123f may be provided in plural. The plurality of mounts 123e and 123f may include a first mount 123e and a second mount 123f.

The first mount 123e may allow the second mover 121b to communicate with the mixing path 222. The first mount 123e may form an outlet 123a. The detergent contained in the detergent pump 120 may flow into the mixing path 222 through the outlet 123a. The first mount 123e may protrude

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forward from the second mount 123f. However, the shape of the first mount 123e is not limited to the example shown in the drawing. The first mount 123e may allow the mounting member 123 to be mounted on the mixing path 222. The first mount 123e may protect the second mover 121b and cover the second mover 121b. The first mount 123e may serve as a cover 123e.

The second mount 123f may be formed near the first mount 123e. The second mount 123f may allow the mounting member 123 to be mounted on the pump 120a. Ends of the second mount 123f may extend toward the pump 120a to be mounted on the pump 120a.

FIG. 11 is a cross-sectional view of the detergent supply device and the circulation device illustrated in FIG. 8. FIG. 12 is a cross-sectional view of the detergent supply device and the circulation device illustrated in FIG. 8. FIG. 11 is a cross-section view of the detergent supply device and the circulation device taken along line A-A' of FIG. 8. FIG. 12 is a cross-sectional view of the detergent supply device and the circulation device taken along line B-B' of FIG. 8.

Referring to FIGS. 11 and 12, the detergent supply device 100 according to an embodiment of the present disclosure may include a detergent case 110, a barrier wall 166 partitioning a plurality of detergent cases 110, an accommodating space 110a formed in the detergent case 110, a detergent pump 120 configured to suck a detergent contained in the accommodating space 110a and discharge the detergent into a mixing space 222a in response to docking with the detergent case 110, a housing 160 accommodating the detergent case 110 and allowing the detergent case 110 to slide in a forward/backward direction, and a case (second drawer) accommodating the detergent case 110 and the housing 160.

In addition, the detergent pump 120 may include a pump 120a, and a mounting member 123 coupled to the pump 120a in front of the pump 120a and mounted on the mixing path 222. The mounting member 123 may be located on the mixing path 222. Therefore, the detergent passing through the interior 123b of the mounting member 123 may immediately flow into the mixing space 222a where wash water flows, and thus an amount of the detergent remaining between the detergent pump 120 and the mixing path 222 may be minimized.

The detergent pump 120 may include a connector 127 electrically and/or physically connected to a device outside the detergent pump 120. The connector 127 may be disposed behind the pump 120a.

In addition, the circulation device 200 according to an embodiment of the present disclosure may include the mixing path 222 constituting the mixing space 222a.

Referring to FIGS. 11 and 12, a flow path of the detergent contained in the accommodating space 110a of the detergent case 110 will be described.

The accommodating space 110a may store the detergent. In response to docking of the detergent case 110 with the detergent pump 120 or the operation of the detergent pump 120, the detergent may be introduced into the pump 120a through the mover 121. For example, the detergent may be introduced into the flow path 121c via the inlet 121d and enter the pump 120a by pushing a first valve 125. The detergent introduced into the pump 120a may be discharged out of the pump 120a by pushing a second valve 126. The detergent discharged from the pump 120a may flow into the mixing space 222a via the outlet 123a formed at one end of the mounting member 123 from the interior 123b of the mounting member 123. The outlet 123a may communicate

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with the mixing space 222a. For example, the outlet 123a may be connected to the inlet 222d.

Because wash water circulating in the circulation device 200 pass through the mixing space 222a, the detergent may be immediately mixed with wash water. Wash water mixed with the detergent, i.e., mixed water, may flow into the tub interior 20c through the circulation pump 210 and the second circulation path 230.

FIG. 13 is a control block diagram of the clothes treating apparatus illustrated in FIG. 1. FIG. 14 is a control flow chart of the clothes treating apparatus illustrated in FIG. 1.

Referring to FIGS. 13 and 14, the washing machine according to an embodiment of the present disclosure may include a processor 300. The processor 300 may control the circulation pump 210, the detergent pump 120, and the drain pump 240. For example, the processor 300 may turn on/off the circulation pump 210, the detergent pump 120, and/or the drain pump 240. In addition, the processor 300 may optionally operate the circulation pump 210 or the drain pump 240. For example, the processor 300 may turn on the circulation pump 210 and turn off the drain pump 240. On the contrary, the processor 300 may turn off the circulation pump 210 and turn on the drain pump 240.

Referring to FIGS. 7 to 14, a sequence of controlling the operation of the circulation device 200 in the washing machine according to an embodiment of the present disclosure will be described.

Water (wash water) may be supplied to the washing machine 1 from the outside through the water supply device 14. The water supply device 14 may supply wash water into the interior 20c of the tub 20 (1410). In this case, because flow paths connected to the tub 20 are closed, water may be collected in the tub 20. The processor 300 may sense a water level of the tub interior 20c using the water level sensor 250 provided at an upper portion of the tub 20. In the case where the water level of the tub interior 20c reaches a predetermined water level, the circulation pump 210 may operate (1420). For example, upon determination that the water level of the tub interior 20c reaches $\frac{1}{3}$ or more of a height of the tub 20, the processor 300 may operate the circulation pump 210 (1430). In response to the operation of the circulation pump 210, water contained in the tub interior 20c may flow into the circulation device 200 disposed below the tub 20. For example, upon the operation of the circulation pump 210, water contained in the tub interior 20c may be guided through the drain path 221 to flow toward the mixing path 222.

For example, wash water may be discharged from the tub 20 through the first drain path 221a and guided to the second drain path 221b. Wash water guided to the second drain path 221b may be guided to the mixing path 222.

In response to the operation of the circulation pump 210, the detergent pump 120 may operate (1440). The detergent pump 120 may suck the detergent from the detergent case 110 and migrate the detergent to the mixing space 222a formed in the mixing path 222 (See FIGS. 11 and 12). For example, the detergent may be discharged into the mixing space 222a of the mixing path 222 from the detergent pump 120. Accordingly, wash water passing through the mixing space 222a may be mixed with the detergent discharged from the detergent pump 120.

Wash water mixed with the detergent may be guided to the circulation pump 210. The wash water may flow to the second circulation path 230 after passing through the circulation pump 210. Because the second circulation path 230 is connected to the first circulation path 220 and the tub 20,

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respectively, wash water that has passed through the first circulation path 220 and the circulation pump 210 may flow to the tub interior 20c.

On the contrary, the processor 300 may also operate the drain pump 240 without operating the circulation pump 210. As described in FIG. 7, in response to the operation of the drain pump 240, wash water of the tub interior 20c may flow into the inner flow path 224 through the first circulation path 220 or the drain holes 223 of the drain path 221.

Wash water may be discharged from the tub 20 through the first drain path 221a and guided to the second drain path 221b. The drain pump 240 may serve as a valve opening or closing the inner flow path 224, and the inner flow path 224 may be open or closed by operating the drain pump 240. In response to the opening of the inner flow path 224, wash water guided to the second drain path 221b may be guided out of the washing machine 1 through the drain pipe 241.

Meanwhile, the washing machine may include a plurality of detergent cases 111 and 112. In addition, the washing machine may include a plurality of modes including a washing mode, a spin-drying mode, a rinsing mode, a drying mode, and the like.

In response to the operation of the circulation pump 210, the processor 300 may operate the detergent pump 120 such that the detergent contained in the first detergent case 111 flows to the mixing space 222a through the detergent pump 120. Thereby, a washing mode may be performed in the tub 20 using mixed water in which the detergent and wash water are mixed. Thereafter, for example, in a final rinsing mode of the washing machine, the processor 300 may operate the detergent pump 120 such that the detergent contained in the second detergent case 112 to flow to the mixing space 222a through the detergent pump 120. Thereby, the rinsing mode may be performed using the mixed water in which the detergent is mixed with wash water. In this case, the detergent contained in the first detergent case 111 may be a laundry detergent, and the detergent contained in the second detergent case 112 may be a fabric softener.

The embodiments of the present disclosure have been shown and described above with reference to the accompanying drawings. However, the disclosed embodiments are illustrative and the scope of the present disclosure is not limited thereby. It will be understood by those of ordinary skill in the art that the present disclosure may be easily modified into other detailed forms without changing the technical principle or essential features of the present disclosure.

What is claimed is:

1. A clothes treating apparatus comprising:

- a cabinet;
- a tub in the cabinet and configured to collect wash water;
- a circulation device configured to circulate the wash water contained in the tub, and the circulation device comprising:
 - a circulation pump;
 - a first circulation path to guide the wash water from the tub to the circulation pump; and
 - a second circulation path to guide the wash water from the circulation pump to the tub so that the wash water circulates by flowing from the tub to the circulation pump through the first circulation path and flowing back to the tub from the circulation pump through the second circulation path; and
- a detergent supply device configured to supply a detergent to the circulation device, the detergent supply device comprising:
 - a detergent case to store the detergent; and

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a detergent pump connected to the detergent case and directly mounted on the first circulation path to supply the stored detergent from the detergent case to the first circulation path, the detergent pump including:

- an inlet through which the detergent is introduced from the detergent case; and
- an outlet through which the introduced detergent is discharged,

wherein the first circulation path comprises a mixing path mounted with the detergent pump, the mixing path including an inlet which is directly connected to the outlet of the detergent pump to receive the detergent from the detergent case thereby mixing the detergent with the wash water flowing in the first circulation path.

2. The clothes treating apparatus according to claim 1, wherein the first circulation path further comprises:

- a drain path to guide the wash water discharged from the tub therethrough.

3. The clothes treating apparatus according to claim 2, wherein the drain path comprises:

- a first drain path to guide the wash water from the tub out of the tub therethrough; and
- a second drain path connected to the first drain path and branching from the first drain path.

4. The clothes treating apparatus according to claim 3, further comprising a drain pump configured to receive the wash water from the first drain path and mounted on the second drain path.

5. The clothes treating apparatus according to claim 4, further comprising a water supply device configured to supply the wash water into the tub from an external water supply source,

- wherein the wash water flowing into the tub by the water supply device flows to the tub after passing through the first circulation path, the circulation pump, and the second circulation path in response to the circulating of the wash water by the circulation pump.

6. The clothes treating apparatus according to claim 5, further comprising a processor configured to selectively operate the circulation pump or the drain pump.

7. The clothes treating apparatus according to claim 6, wherein the processor is further configured to operate the circulation pump in response to determining that a water level in the tub reaches a predetermined water level.

8. The clothes treating apparatus according to claim 7, wherein the processor is further configured to operate the detergent pump in response to the operating of the circulation pump.

9. The clothes treating apparatus according to claim 8, wherein the detergent case comprises:

- a first detergent case; and
- a second detergent case separated from the first detergent case in a lateral direction,

wherein the processor is further configured:

- to operate the circulation pump based on an initiation of a washing operation;
- to operate the detergent pump to allow a detergent contained in the first detergent case to flow into the detergent pump based on the operation of the circulation pump; and
- to operate the detergent pump to allow a detergent contained in the second detergent case to flow into the detergent pump based on an initiation of a rinsing mode.

10. The clothes treating apparatus according to claim 6, wherein the processor is further configured to turn off the

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circulation pump and turn on the drain pump to drain the wash water contained in the tub.

11. The clothes treating apparatus according to claim 5, wherein the detergent supply device and the circulation device are disposed below the tub.

12. The clothes treating apparatus according to claim 11, wherein the detergent supply device is a first detergent supply device, and

- the clothes treating apparatus further comprises a second detergent supply device disposed above the tub to supply the detergent into the tub.

13. The clothes treating apparatus according to claim 12, further comprising:

- a fan above the tub to form a circulating air flow passing through the tub; and
- a heat exchanger above the tub to dehumidify and heat air discharged from the tub and supply the dehumidified and heated air to the tub.

14. The clothes treating apparatus according to claim 1, wherein the detergent supply device further comprises a drawer configured to accommodate the detergent case and to be withdrawn from or inserted into the cabinet.

15. The clothes treating apparatus according to claim 14, wherein the drawer comprises:

- a first drawer configured to accommodate the detergent case and withdrawn from or inserted into the cabinet; and
- a second drawer coupled to a side wall of the cabinet, configured to accommodate the first drawer, and slidably coupled to the first drawer allowing the first drawer to slide.

16. The clothes treating apparatus of claim 1, wherein the mixing path comprises a mixing space formed therein to receive the wash water from the tub and the discharged detergent through the inlet of the mixing path which is directly connected to the outlet of the detergent pump.

17. The clothes treating apparatus of claim 16, wherein the supplied wash water from an external water supply source flows back to the tub after passing through the water supply device, the tub, and the mixing space.

18. A clothes treating apparatus comprising:

- a cabinet;
- a tub in the cabinet and configured to collect wash water;
- a detergent supply device configured to supply a detergent into the tub, the detergent supply device including:
- a detergent case to accommodate the detergent;
- a detergent pump including:
- an inlet through which the detergent is introduced from the detergent case; and
- an outlet through which the introduced detergent is discharged; and
- a mixing space formed therein to receive the wash water from the tub and directly connected to the outlet of the detergent pump to receive the discharged detergent from the detergent pump;
- a drain pump connected to the tub to drain the wash water contained in the tub out of the cabinet;
- a circulation pump connected to the tub to circulate the wash water;
- a first circulation path to guide the wash water from the tub to the circulation pump;
- a second circulation path to guide the wash water from the circulation pump to the tub so that the wash water circulates by flowing from the tub to the circulation pump through the first circulation path and flowing back to the tub from the circulation pump through the second circulation path; and

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a processor configured to selectively operate the drain pump or the circulation pump, wherein the first circulation path includes a mixing path on which the outlet of the detergent pump is mounted and having a mixing space formed therein to receive the wash water from the tub and directly connected to the outlet of the detergent pump to receive the discharged detergent from the detergent pump. 5

19. The clothes treating apparatus of claim **18**, wherein the processor is further configured to operate the detergent pump in response to the operating of the circulation pump. 10

20. The clothes treating apparatus of claim **19**, wherein the processor is further configured to turn off the circulation pump and turn on the drain pump to drain the wash water contained in the tub. 15

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